ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main + Advanced) : ENTHUSIAST COURSE

SCORE - I

Test Type : FULL SYLLABUS Test Pattern : JEE-Advanced

TEST DATE : 16 - 06 - 2021

Time : 3 Hours **PAPER – 1** **Maximum Marks : 186**

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble ◯ **COMPLETELY.**
12. The correct way of darkening a bubble is as : ●
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **g = 10 m/s²** unless otherwise stated.

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS	
Atomic No.	H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58,
Atomic masses :	H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca=40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba=137, Ce = 140,

• **Boltzmann constant**	$k = 1.38 \times 10^{-23}\ JK^{-1}$
• **Coulomb's law constant**	$\frac{1}{4\pi\varepsilon_0} = 9\times10^{9}$
• **Universal gravitational constant**	$G = 6.67259 \times 10^{-11}\ N\text{–}m^2\ kg^{-2}$
• **Speed of light in vacuum**	$c = 3 \times 10^{8}\ ms^{-1}$
• **Stefan–Boltzmann constant**	$\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\text{–}K^{-4}$
• **Wien's displacement law constant**	$b = 2.89 \times 10^{-3}\ m\text{–}K$
• **Permeability of vacuum**	$\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$
• **Permittivity of vacuum**	$\epsilon_0 = \frac{1}{\mu_0 c^2}$
• **Planck constant**	$h = 6.63 \times 10^{-34}\ J\text{–}s$

Space for Rough Work

HAVE CONTROL ⟶ HAVE PATIENCE ⟶ HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

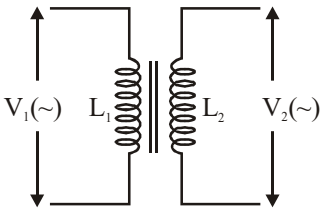
SECTION–I(i) : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases

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1. For an ideal transformer, ratio of V_1 & V_2 is equal to



Where L_1 & L_2 are self inductances of primary and secondary windings. (For an ideal transformer coefficient of mutual induction for pair of coils is $M = \sqrt{L_1 L_2}$) :-

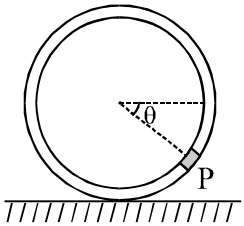
(A) $\frac{L_1}{L_2}$ (B) $\sqrt{\frac{L_1}{L_2}}$ (C) $\frac{L_2}{L_1}$ (D) $\sqrt{\frac{L_2}{L_1}}$

Space for Rough Work

2. According to Sommerfield model of atom an electron revolve around nucleus in an elliptical orbit while nucleus remain at rest. If minimum and maximum distance of electron from nucleus is r_1 & r_2 then find distance of electron from nucleus when it is at a position where line joining electron & nucleus is perpendicular to major axis :-

(A) $r_1 + r_2$ (B) $r_2 - r_1$ (C) $\frac{r_1 r_2}{r_1 + r_2}$ (D) $\frac{2r_1 r_2}{r_1 + r_2}$

3. A small block of mass m is rigidly attached at 'P' to a ring of mass m and radius r. The system is released from rest at $\theta = 30°$ and rolls without sliding. The speed of centre of mass of system when block reaches the bottom is



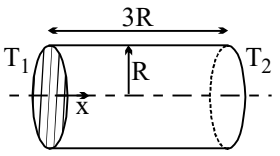
(A) $\sqrt{rg}$ (B) $\frac{\sqrt{rg}}{2}$ (C) $\sqrt{\frac{rg}{2}}$ (D) none

Space for Rough Work

4. One mole of an ideal monoatomic gas is enclosed in a chamber at 300 K. The gas undergoes a process in which pressure is proportional to the volume. At the end of the process, the volume of the gas is doubled. The change in the internal energy of the gas is [R is gas constant]

(A) 450 R (B) 700 R (C) 1350 R (D) data insufficient

5. For a conducting cylinder with variable thermal conductivity along the axis the magnitude of rate of heat flow if temperature difference across ends is $(T_1 - T_2)$,



solid cylinder, flow along axis,
variable k as $k = k_0(1+x/(3R))$
x is distance from left end

(A) $6\pi k_0 R(T_1 - T_2)$ (B) $\frac{\pi k_0 R}{3\ln 2}(T_1 - T_2)$ (C) $\pi k_0 R(T_1 - T_2)$ (D) $\frac{4\pi k_0 R}{\ln 2}(T_1 - T_2)$

Space for Rough Work

SECTION–I(ii) : (Maximum Marks : 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- For each question, marks will be awarded in one of the following categories :

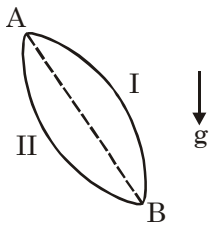
  Full Marks : +4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened.

  Partial Marks : +1 For darkening a bubble corresponding **to each correct option**, Provided NO incorrect option is darkened.

  Zero Marks : 0 If none of the bubbles is darkened.

  Negative Marks : –2 In all other cases.
- for example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in +4 marks; darkening only (A) and (D) will result in +2 marks; and darkening (A) and (B) will result in –2 marks, as a wrong option is also darkened

6. A small bob can slide downwards from point A to B along either of the two different curved surfaces shown in cross-section in the diagram. These possible trajectories are circular arcs in vertical plane, and they lie symmetrically about straight line AB. During the motion, bob does not leave contact with surface.



(A) If friction is neglected every where bob will take a smaller time through path I as compared to time through path II.

(B) If friction is neglected every where bob will take a smaller time through path II as compared to time through path I.

(C) If friction is neglected every where bob will reach point B with same speed through both the paths.

(D) If friction is significant along both paths then bob will reach B with a smaller speed, if it follows path II, as compared to if it follows path I.

Space for Rough Work

7. On a spherical planet, the refractive index of atmosphere, as a function of altitude h above the surface varies as

$$n(h)=\frac{n_0}{1+\frac{h}{2R}}$$

Where n_0 is a constant. A laser beam directed horizontally but an arbitrary altitude follows a trajectory that circles the planet.

(A) Radius of planet has a magnitude $R\sqrt{2}$

(B) Radius of planet has a magnitude 2R

(C) At a height $R\sqrt{2}$ from the surface of planet, refractive index is $\frac{n_0}{2}$

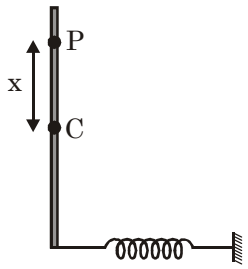
(D) At a height R from surface, refractive index is $\frac{2n_0}{3}$

8. Two equal masses m are at a distance ℓ apart and interact via gravity. They are given proper tangential speed v_0, so that they both travel in a circular of radius $\frac{\ell}{2}$ around their CM. If one of the masses is grabbed and held at rest, then the closest distance the other mass comes to it is r. Then :

(A) $v_0=\sqrt{\frac{Gm}{2\ell}}$ (B) $v_0=\sqrt{\frac{Gm}{\ell}}$ (C) $r=\frac{\ell}{3}$ (D) $r=3\ell$

Space for Rough Work

9. A uniform stick with mass m and length ℓ lies on a frictionless horizontal surface. It is pivoted at a point a distance x from centre. A spring (at its relaxed length) with spring constant k is attached to the far end of the stick, perpendicular to the stick, as shown. Stick is given a small kick, then :



(A) Value of x which yields largest frequency of oscillation is $\frac{\ell}{3}$.

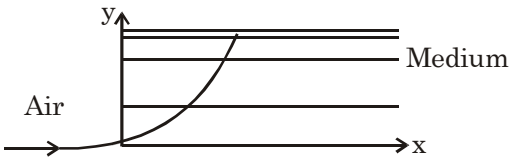
(B) Value of x which yields largest frequency of oscillation is $\frac{\ell}{6}$.

(C) Angular frequency of oscillation (largest value) is $\frac{2k}{m}$.

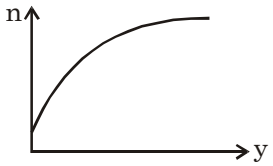
(D) Angular frequency of oscillation (largest value) is $\frac{4k}{m}$.

Space for Rough Work

10. The refractive index of the medium within a certain region x > 0, y > 0, changes continuously with y. A thin light ray travelling in air in the x-direction strikes the medium at right angles and moves through the medium along a circular arc of radius R.



(A) Refractive index of medium varies with y as



(B) Refractive index of medium varies with y as



(C) If refractive index of medium can increase upto a value n = 2.5, the maximum value of y is $\frac{3R}{5}$

(D) If refractive index of medium can increase upto a value n = 2.5, the maximum value of y is 5R

Space for Rough Work

11. Two waves given by equation $y_1 = A \sin(\omega t - kx)$ and $y_2 = 4A \sin(2\omega t - 4kx)$ are traveling in two different strings having same mass per unit length and same cross-sectional area. Tension in first string is four times tension in the second string.

(A) Ratio of average energy density U_1/U_2 is 1/64

(B) Ratio of average intensity I_1/I_2 is 1/32

(C) Ratio of average power P_1/P_2 is 1/16

(D) Ratio of velocity of wave v_1/v_2 is 1/2

12. A metal block is placed in a room which is at 10°C for long time. Now it is heated by an electric heater of power 500 W till its temperature becomes 50°C. Its initial rate of rise of temperature is 2.5°C/sec. The heater is switched off and now a heater of 100W is required to maintain the temperature of the block at 50°C.(Assume Newtons Law of cooling to be valid)

(A) The heat capacity of the block is 200 J/°C

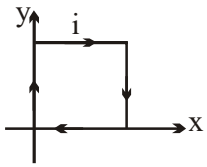
(B) The heat capacity of the block is 100 J/°C

(C) The rate of cooling of block at 50°C if the 100W heater is also switched off is 0.5°C/s

(D) The heat radiated per second when the block was 30°C is 50 W

Space for Rough Work

13. Figure shows a square current carrying coil of edge length L. The magnetic field on the coil is given by $\vec{B}=\frac{B_0 y}{L}\hat{i}+\frac{B_0 x}{L}\hat{j}$ where B_0 is a positive constant. (A is area of coil)



(A) If coil is free to rotate about x axis torque on the coil is given by $\frac{1}{2}iAB_0\hat{i}$.

(B) If coil is free to rotate about y - axis torque on coil is given by $-\frac{1}{2}iAB_0\hat{j}$.

(C) Resultant force on coil is zero.

(D) Equation for the torque $\vec{\mu}\times\vec{B}$ where μ is magnetic moment of coil is not valid on the coil if any of the side is fixed as axis.

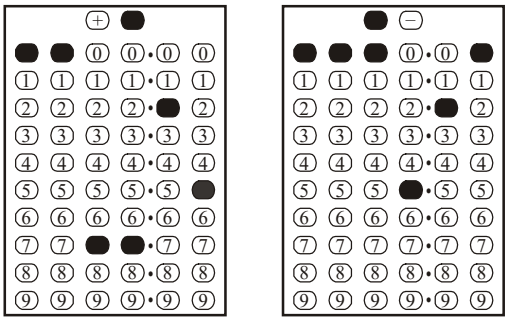
Space for Rough Work

SECTION –II : Matrix-Match Type & SECTION –III : Integer Value Correct Type
No question will be asked in section II and III

SECTION–IV : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

  For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.

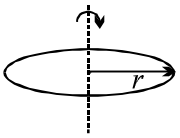


- Answer to each question will be evaluated according to the following marking scheme:

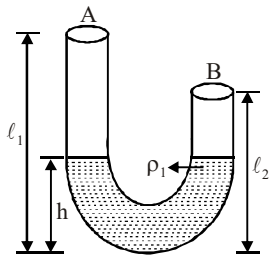
  Full Marks : +3 If ONLY the correct numerical value is entered as answer.

  Zero Marks : 0 In all other cases.

1. A ring of radius r made of wire of density ρ is rotated about a stationary vertical axis passing through its centre and perpendicular to the plane of the ring as shown in figure. Find the percentage change in tension for 3 percent change in angular velocity of rotation. Ignore gravity.



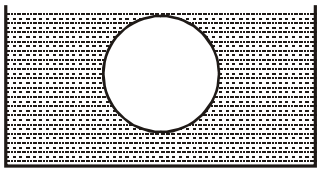
2. A U-tube having uniform cross-section but unequal arm length l_1 = 100 cm and l_2 = 50 cm has same liquid of density ρ_1 filled in it upto a height h = 30 cm as shown in figure. Another liquid of density $\rho_2 = 2\rho_1$ is poured in arm A. Both liquids are immiscible. What length of the second liquid (in cm) should be poured in A so that second overtone of A is in unison with fundamental tone of B. (Neglect end correction)



Space for Rough Work

3. A source of frequency 175 Hz is placed between a man and a wall at the same height of person's ear. Find the velocity v (assume v << c) of the source with which it should approach the wall such that person will detect 3 beats per second. [Take velocity of sound c = 350 m/s]

4. A sphere is floating in a liquid in just submerged position as shown. Find the ratio of magnitude of forces exerted by liquid on lower half of sphere to that exerted by liquid on upper half of sphere (Neglect atmospheric pressure)



5. One mole of an ideal monoatomic gas is taken from state A to state B through the process $P = \frac{3}{2}T^{1/2}$. It is found that its temperature increases by 100 K in this process. Now it is taken from state B to C through a process for which internal energy is related to volume as $U = \frac{1}{2}V^{1/2}$. The total work performed by the gas (in Joule) is W, if it is given that volume at B is 100 m^3 and at C it is 1600 m^3. [Use R = 8.3 J/mol-K]. Fill W/87.

Space for Rough Work

PART-2 : CHEMISTRY

SECTION–I(i) : (Maximum Marks : 15)

- This section contains **FIVE** questions.
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- For each question, darken the bubble corresponding to the correct option in the ORS.
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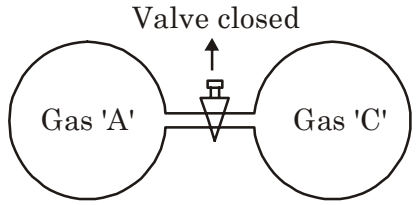
1. Which one of the following dimeric form of compounds is planar ?

(A) B_2H_6 (B) Al_2Cl_6 (C) Al_2Br_6 (D) I_2Cl_6

2. Borazine + $H_2O \longrightarrow H_3BO_3 + A_{(g)} + B_{(g)}$

$B + N_2 \longrightarrow A$; $B + Cl_2 \longrightarrow C$

$A_{(g)} + C_{(g)} \longrightarrow D_{(s)}$



Flask - I T = 300 K P = 485 mm Hg

Flask - II T = 400 K P = 600 mm Hg

Two flasks – I & II of equal volume containing Gas 'A' and Gas 'C', are connected by a narrow tube of negligible volume. The two gases were prevented from mixing by stopper fitted in connecting tube. For further detail of experiment refer to the given figure. What will be final pressure in each flask when valve connecting two tubes is opened. Assume ideal gas behaviour.

(A) 40 mm Hg (B) 60 mm Hg (C) 20 mm Hg (D) 10 mm Hg

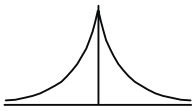
3. A single walled carbon nano–tube can be regarded an a single sheet of graphite rolled into a cylinder. Number of C–atom in a single walled nano–tube having length 1.0 mm & diameter 1.08 nm ?

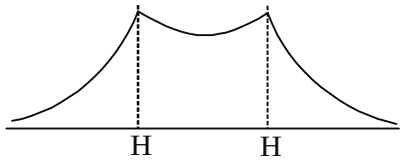
[Given : Carbon - Carbon bond length in graphite = 141.5 pm]

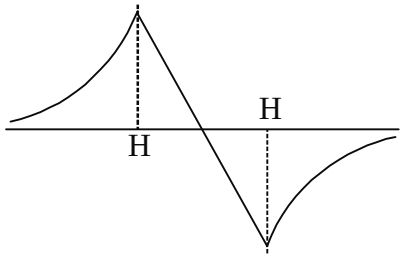
(A) 0.65×10^8 (B) 2.6×10^8 (C) 3.9×10^8 (D) 1.3×10^8

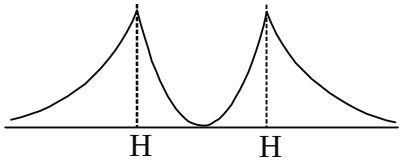
Space for Rough Work

4. For the interaction H + H $\longrightarrow$ H_2, **INCORRECT** is –

(A) Ψ_H vs r 

(B) $(\Psi_{bonding})_{H_2}$ vs r 

(C) $(\Psi_{antibonding})_{H_2}$ vs r 

(D) $(\Psi_{antibonding})_{H_2}$ vs r 

5. Identify the **CORRECT** option.

(I) AgCl is not soluble in H_2O but soluble in KCN.

(II) Zn when reacts with very dil HNO_3(6%) gives NH_4NO_3.

(III) $CuSO_4$ reacts with KI to form only Cu_2I_2 + K_2SO_4.

(IV) $CuSO_4.5H_2O \xrightarrow{1000^oC}$ The gases liberated are SO_2 and O_2.

(A) All statements are correct (B) I, II and IV are correct

(C) II and III are incorrect statements (D) Only I is incorrect.

Space for Rough Work

SECTION–I(ii) : (Maximum Marks : 32)

- This section contains **EIGHT** questions.
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Zero Marks : 0 If none of the bubbles is darkened.

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- for example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in +4 marks; darkening only (A) and (D) will result in +2 marks; and darkening (A) and (B) will result in –2 marks, as a wrong option is also darkened

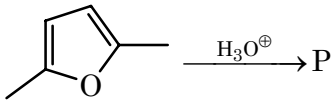
6. Which of the following options is **CORRECT** ?

(A) Tailing of mercury related to test of Ozone.

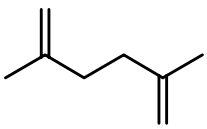
(B) Carbon suboxide (C_3O_2) is linear molecule and prepared by dehydration of Malonic acid

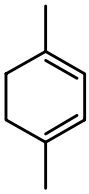
(C) K_2O and K_2O_2 are diamagnetic but KO_2 and KO_3 are paramagnetic

(D) Mn_2O_7 is acidic oxide but MnO is amphoteric

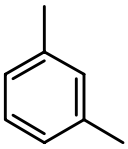
7. 

An organic compound on reaction with O_3 followed by Zn and H_2O gives P. The organic compound can be ?

(A) 

(B) 

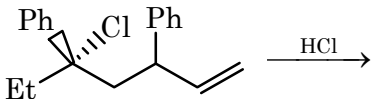
(C) 

(D) 

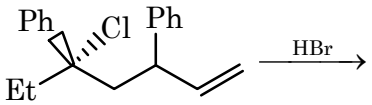
Space for Rough Work

8. Select correct statement for following reactions :

Reaction : I



Reaction : II



(A) Reaction I gives mixture of 2 enantiomers & Reaction II gives mixture of two diastereomers.

(B) Both reactions give mixture of 2 enantiomers.

(C) Both reactions give mixture of 2 diastereomers.

(D) Both reactions give 2 fraction of organic compounds.

9. $CH_3—CH=CH—\underset{\displaystyle H}{\overset{|}{C}}=O$ (excess) $\xrightarrow[\Delta]{LDA}$ A $\xrightarrow[\text{K (excess)}]{\text{THF, 60°C, 15 bar}}$ B, C

(A) 'A' is anti–aromatic.

(B) 'B' is aromatic.

(C) 'A' is having degree of unsaturation = 5.

(D) In the compound 'C', C–H bond length is more than C–C bond length.

10. Find **CORRECT** statements.

(A) Diamond is better conductor of heat than graphite.

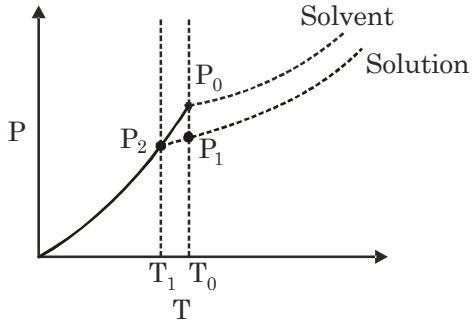
(B) Diamond is better conductor of electricity than Ag.

(C) Diamond is electrical insulater.

(D) Diamond gets converted to graphite at very high pressure & temperature.

Space for Rough Work

11. From the phase diagram of water and an aqueous solution containing non volatile solute, identify the **correct** options:



(A) At temperature T_0, vapour pressure of solid solvent and vapour pressure of liquid solvent will be same.

(B) Order of vapour pressure (P_0, P_1, P_2) are $P_0 > P_1 > P_2$.

(C) $P_0 = P_2 e^{\frac{\Delta H_{fusion}}{R}\frac{[T_0 - T_1]}{T_0 T_1}}$

(D) $P_1 = P_2 e^{\frac{\Delta H_{vap}}{R}\frac{[T_0 - T_1]}{T_0 T_1}}$

12. Which of the following statements are correct ?

(A) $\frac{k_p}{k_a} = e^{\Delta E/RT}$ where k_p = Rate constant for reaction with catalyst.
K_a = Rate constant for reaction without catalyst
ΔE = Difference in activation energy for the two ways of reactions.

(B) Alcohol acts as a negative catalyst for the conversion of $Na_2SO_3 \longrightarrow Na_2SO_4$.

(C) TEL (tetraethyl lead) acts as a negative catalyst in the knocking of petrol.

(D) Dilute H_2SO_4 acts as a positive catalyst in the hydrolysis of esters to carboxylic acids and alcohols.

13. In which of the following cases, the synergic bonding takes place at the π^*-orbital of the ligand.

(A) $[PtCl_3(C_2H_4)]^-$ (B) $[Ni(PF_3)_4]$ (C) $[Cr(C_6H_6)_2]$ (D) $[Fe(\pi - C_5H_5)_2]$

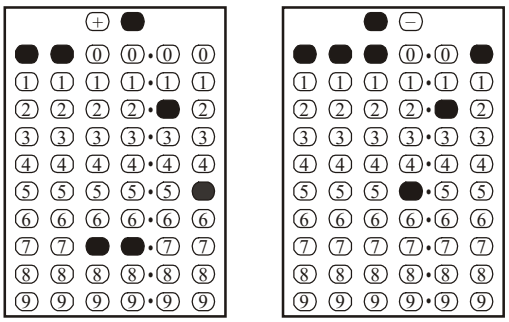
Space for Rough Work

SECTION –II : Matrix-Match Type & SECTION –III : Integer Value Correct Type
No question will be asked in section II and III

SECTION–IV : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

  For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



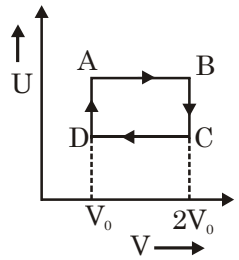
- Answer to each question will be evaluated according to the following marking scheme:

  Full Marks : +3 If ONLY the correct numerical value is entered as answer.

  Zero Marks : 0 In all other cases.

---

1. The internal energy (U) of an ideal (10.7 mole) gas is plotted against volume for a cyclic process ABCDA, as shown in the figure.



The temperature of the gas at B and C are 500 K and 300 K, respectively. The heat absorbed by the gas (in kcal) in this cyclic process is :

2. E°_{Cell} for the cell

$$Pt(s)\,|\,Fe^{2+}(aq)\,|\,Fe^{3+}(aq)\,||\,MnO_4^-(aq),H^+(aq)\,|\,Mn^{2+}(aq)\,|\,Pt$$ is 0.788 Volt

Given :

$$\Delta G^{\circ}_f H_2O(\ell) = -230\,kJmol^{-1}$$

$$\Delta G^{\circ}_f Fe^{3+}(aq) = -10.2\,kJmol^{-1}$$

$$\Delta G^{\circ}_f Mn^{2+}(aq) = -229\,kJmol^{-1}$$

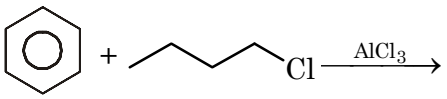
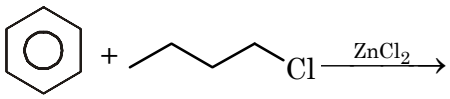
$$\Delta G^{\circ}_f Fe^{2+}(aq) = -84\,kJmol^{-1}$$

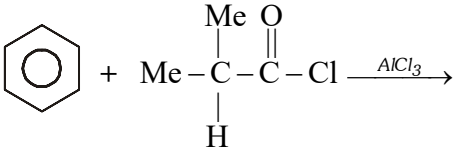
Magnitude of $\Delta G^{\circ}_f MnO_4^-(aq)$ in kJ mol^{-1} is $x \times 10^2$ then 'x' is-

(Give your answer as nearest integer)

Space for Rough Work

3. Three reactions are given below, write sum of total number of hydrogen atoms on all the carbon atoms which are connected directly by a single bond to benzylic carbon (carbon connected to benzene ring) in the products.

(a)  (b) 

(c) 

4. (a) From Meridional and facial isomer of $[Ma_3b_3]^{n\pm}$ on replacement of only one 'a' by 'b', the number of isomer of the product obtained are respectively x and y.

(b) The number of corner shared per tetrahedron for 2D-silicate is z.

Then x + y + z is-

5. An optically active substance 'A' is decomposing into optically active substance 'B' and 'C' as

$$A \xrightarrow{K=10^{-3}\,min^{-1}} 2B + C$$

The specific rotations of A, B and C are + 40° , + 10° and –30° per mole, respectively. Initially 'A' and 'C' were present in 4 : 3 mole ratio. Close to how many hours the sample becomes optically inactive ?

[Given : $\ell n\frac{7}{5} = 0.34, \ell n\frac{20}{13} = 0.42$]

Space for Rough Work

PART-3 : MATHEMATICS

SECTION–I(i) : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases

---

1. Let the function $f : R \to R$ be defined as $f(x) = \min.(x + 2, 4 - 2x, 1 + 4x)$. The maximum value of $f(x)$ is equal to -

(A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{8}{3}$

2. Number of real roots of the equation $\sqrt{1+3x}+\sqrt{3+5x}+\sqrt{5+x}=\sqrt{3-x}+\sqrt{5-3x}+\sqrt{1-5x}$ is -

(A) 1 (B) 2 (C) 3 (D) infinite

3. In a sequence $\{a_n\}, a_1 = 2, a_{n+1} = 1 - \frac{1}{a_n}$ for $n \geq 1$. Let P_n be the product of its first n terms, then the value of P_{2017} is

(A) $\frac{-1}{2}$ (B) 2 (C) $\frac{1}{2}$ (D) 1

4. Given that 'a' is a root of the equation $x^2 - x - 3 = 0$ then $\frac{a^3+1}{a^5-a^4-a^3+a^2}$ equals

(A) $\frac{4}{3}$ (B) $\frac{3}{4}$ (C) $\frac{4}{9}$ (D) $\frac{9}{4}$

5. If $p, q, r \in R$ satisfy $4p + 3q + r = 7$, then the least value of $(2p^2 + q^2 + r^2)$ is equal to -

(A) $\sqrt{7}$ (B) $\frac{7}{\sqrt{2}}$ (C) $\frac{\sqrt{7}}{2}$ (D) $\frac{49}{18}$

Space for Rough Work

SECTION–I(ii) : (Maximum Marks : 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened.

Partial Marks : +1 For darkening a bubble corresponding **to each correct option**, Provided NO incorrect option is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks : –2 In all other cases.

- for example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in +4 marks; darkening only (A) and (D) will result in +2 marks; and darkening (A) and (B) will result in –2 marks, as a wrong option is also darkened

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6. Let us define system of vectors $\vec{u}_1 = \vec{v}_1$ and $\vec{u}_k = \vec{v}_k - \text{proj}_{\vec{u}_1}(\vec{v}_k) - \ldots - \text{proj}_{\vec{u}_{k-1}}(\vec{v}_k), k \geq 2$, where $\text{proj}_{\vec{a}}(\vec{b}) = \dfrac{(\vec{b}\cdot\vec{a})\vec{a}}{|\vec{a}|^2}$ and $\vec{v}_1 = 2\hat{i}+\hat{j}-\hat{k}, \vec{v}_2 = \hat{i}+\hat{j}+\hat{k}, \vec{v}_3 = 3\hat{i}-2\hat{j}-\hat{k}$. Which of the following is correct ?

(A) $\left[\vec{u}_1\ \vec{u}_2\ \vec{u}_3\right] \neq 0$ (B) $\vec{u}_2 \cdot \vec{v}_1 \neq 0$ (C) $\vec{u}_3 \cdot \vec{v}_1 = 0$ (D) $\vec{u}_3 \cdot \vec{v}_2 \neq 0$

7. For a 3 × 3 invertible matrix A satisfying the characteristic equation $A^3 + pA^2 + qA - rI_3 = 0$, which of the following is/are true [Here tr(A) = trace of matrix A, det(A)=determinant value of matrix A]

(A) $p = -\text{tr}(A)$ (B) $q = \dfrac{(\text{tr}(A))^2 - \text{tr}(A^2)}{2}$

(C) $r = det(A)$ (D) $A^{-1} = \dfrac{-2(A - \text{tr}(A)I_3)}{(\text{tr}(A))^2 - \text{tr}(A^2)}$

Space for Rough Work

8. Let $L_1 : \frac{x+3}{-4} = \frac{y-6}{3} = \frac{z}{2}$ and $L_2 : \frac{x-2}{-4} = \frac{y+1}{1} = \frac{z-6}{1}$ be two lines in R^3. Which of the following statement(s) is(are) correct ?

(A) L_1, L_2 are coplanar.

(B) L_1, L_2 are skew lines.

(C) shortest distance between L_1 and L_2 is 9.

(D) $(\hat{i} - 4\hat{j} + 8\hat{k})$ is a vector perpendicular to both L_1 and L_2.

9. If $(x+1)(x+2)\ .....(x+99) = A_0 + A_1 \cdot x + A_2 \cdot x^2 + ... + A_{99} \cdot x^{99}$ and

$S = \frac{\sum_{k=1}^{99} (k.A_k)}{100!} = \sum_{i=1}^{99} \left(\frac{1}{r_i}\right), \left(r_i < r_j, \forall i < j\right)$ then value of $\sum_{k=1}^{99} \frac{1}{\sqrt{(r_k - 1)}}$ is lying between

(A) 15 and 19 (B) 15 and 17

(C) 14 and 17 (D) 17 and 19

10. Volume of parallelopiped formed by planes $2x + 2y + z = 4$, $2x + 2y + z = 1$, $2x - y + z = 3$, $2x - y + z = 6$, $3x + 4y + 12z = -1$, $3x + 4y + 12z = 12$ is $\frac{p}{q}$ where p and q are positive coprime integers then :

(A) $p + q = 20$ (B) $p + q = 57$

(C) $p - q = 6$ (D) $p - q = 5$

Space for Rough Work

11. A line divides a plane into 2 regions. Two lines divide the plane into maximum 4 regions. If L_n is the maximum number of regions divided by n lines then which of the following is/are true :
(A) $L_{20} = 211$ (B) $L_{10} = 56$ (C) $L_{15} = 121$ (D) $L_{25} = 326$

12. Let $A(z_1)$ be the point of intersection of curves $\arg(z - 2 + i) = \frac{3\pi}{4}$ and $\arg(z + i\sqrt{3}) = \frac{\pi}{3}$. $B(z_2)$ be the point on the curve $\arg\left(z + i\sqrt{3}\right) = \frac{\pi}{3}$ such that $|z_2 - 5|$ is minimum and $C(z_3)$ be the centre of circle $|z - 5| = 3$. [Note : $i^2 = -1$].
Then which of the following statement(s) is (are) correct ?
(A) Distance between $A(z_1)$ and $B(z_2)$ is equal to 3.
(B) Distance between $A(z_1)$ and $B(z_2)$ is equal to 4.
(C) Area of triangle ABC is equal to $3\sqrt{2}$.
(D) Area of triangle ABC is equal to $2\sqrt{3}$.

13. Given that the real numbers s, t satisfy $19s^2 + 99s + 1 = 0$, $t^2 + 99t + 19 = 0$ and $st \neq 1$. The absolute value of $\frac{st + 4s + 1}{t}$, is
(A) a prime number (B) divisible by 19
(C) divisible by 5 (D) divisible by 2

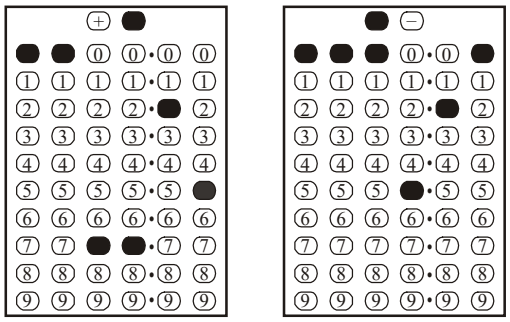
Space for Rough Work

SECTION –II : Matrix-Match Type & SECTION –III : Integer Value Correct Type
No question will be asked in section II and III

SECTION–IV : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

  For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:

  Full Marks : +3 If ONLY the correct numerical value is entered as answer.

  Zero Marks : 0 In all other cases.

---

1. Consider the system of equations $ax + y = b$, $bx + y = a$, $ax + by = ab$, where $a, b \in \{0, 1, 2, 3, 4\}$. The number of ordered pairs (a, b) for which system is consistent is equal to

2. Let $A = \begin{bmatrix} 4 & a \\ -1 & b \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}$ be two matrices satisfying the relation,

$A^3 + 3A^2B + 3AB^2 + B^3 = (A + B)^3$. If $(A + B) = \lambda(A^{-1} + B^{-1})$, then λ is equal to

---

Space for Rough Work

3. Let the focus of conic $\frac{(y+x)^2}{16}-\frac{(y-x)^2}{8}=1$ is at (a, b) then $\frac{a^2+b^2}{4}$ is

4. Let z and ω be complex numbers such that $z+\omega=i$ and $z^2+\omega^2=1$. If the area of triangle formed by z, ω and origin is A, then $2\sqrt{80A^2+1}$ is equal to

5. Let $f : R \to R$ be non-constant differentiable function and satisfies, $f(x)=x^2-\int_0^1 (x+f(t))^2 dt$. Then $f(4)$ is equal to

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.
17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	5	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	15
I(ii)	One or more correct option(s)	8	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–2 In all other cases	32
IV	Single digit Integer (0-9)	5	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	15

NAME OF THE CANDIDATE .....

FORM NO. .....

I have read all the instructions and shall abide by them.	I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.
_____ Signature of the Candidate	_____ Signature of the Invigilator

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2021 - 2022)

JEE (Main + Advanced) : ENTHUSIAST COURSE

SCORE - I

Test Type : FULL SYLLABUS Test Pattern : JEE-Advanced

TEST DATE : 16 - 06 - 2021

Time : 3 Hours **PAPER – 2** **Maximum Marks : 186**

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble ◯ **COMPLETELY.**
12. The correct way of darkening a bubble is as : ●
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **g = 10 m/s²** unless otherwise stated.

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No.	H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58,
Atomic masses :	H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg=24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca=40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba=137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23}\ JK^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\varepsilon_0} = 9\times10^{9}$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11}\ N\text{–}m^2\ kg^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^{8}\ ms^{-1}$
- **Stefan–Boltzmann constant** $\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\text{–}K^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3}\ m\text{–}K$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34}\ J\text{–}s$

Space for Rough Work

HAVE CONTROL ⟶ HAVE PATIENCE ⟶ HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

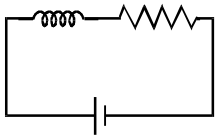
SECTION–I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases

---

1. Time constant of the given circuit is τ. If the battery is replaced by an ac source having voltage $V = V_0 \cos \omega t$, power factor of the circuit will be



(A) $\omega\tau$ (B) $\frac{1}{\sqrt{1+(\omega\tau)^2}}$ (C) $\sqrt{1-(\omega\tau)^2}$ (D) None

2. Coherent light consisting of two wavelengths $\lambda_1 = 4500$ Å & $\lambda_2 = 6000$ Å is sent through both slits of a Young's double slit apparatus (Source is placed symmetrically for both the slits)

(A) Central maxima for both wavelengths will coincide

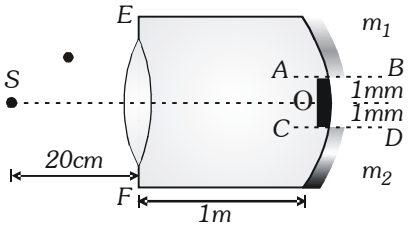
(B) The third order bright fringe of λ_1 will coincide with the fourth order bright fringe of λ_2

(C) The third order dark fringe of λ_2 will coincide with fourth order dark fringe of λ_1

(D) The fringes of wavelength λ_1 will be wider than the fringes of wavelength λ_2

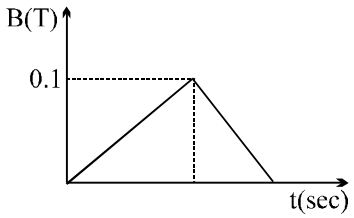
Space for Rough Work

3 An equiconvex lens of focal-length 10 cm (in air) and refractive index 3/2 is put at a small opening on a tube of length 1m fully filled with liquid of refractive index 4/3. A concave mirror of radius of curvature 20 cm is cut into two halves m_1 and m_2 and placed at the end of the tube. m_1 and m_2 are placed such that their principal axes AB and CD respectively are separated by 1mm each from the principle axes of the lens. A slit S placed in air illuminates the lens with light of frequency 7.5×10^{14} Hz(in air). The light reflected from m_1 and m_2 forms interference pattern on the left end EF of the tube taking lens surface as screen. O is an opaque substance to cover the hole left by m_1 and m_2. The width of the fringes on EF.



(A) 180μm (B) 220μm (C) 120μm (D) 60μm

4. A closed loop of cross sectional area 10^{-2} m^2 which has inductance L = 10 mH and negligible resistance is placed in a time varying magnetic field. The figure shows the variation of B with time for the interval 4 sec. The field is perpendicular to the plane of the loop (given at t = 0, B = 0, i = 0). The value of the maximum current induced in the loop is



(A) 0.1 mA (B) 10 mA (C) 100 mA (D) data insufficient

Space for Rough Work

5. A planet of small mass m moves around sun of mass M along an elliptical orbit such that its minimum and maximum distance from sun are r and R respectively. Its period of revolution will be

(A) $2\pi\sqrt{\frac{(r+R)^2}{6GM}}$ (B) $2\pi\sqrt{\frac{(r+R)^3}{3GM}}$ (C) $\pi\sqrt{\frac{(r+R)^3}{2GM}}$ (D) $2\pi\sqrt{\frac{(r+R)^3}{GM}}$

6. A parallel plate capacitor is charged using a battery, and the battery is then removed. The plates of the capacitor are then brought closer together. Which of the following statements is false ?

(A) The electric field inside the capacitor remains the same

(B) The capacitance of the capacitor increases

(C) The potential difference between the plates of the capacitor decreases

(D) The energy stored in the capacitor increases

Space for Rough Work

SECTION–I(ii) : (Maximum Marks : 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened.

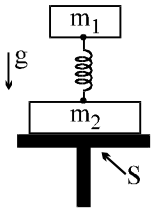
Partial Marks : +1 For darkening a bubble corresponding **to each correct option**, Provided NO incorrect option is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks : –2 In all other cases.

- for example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in +4 marks; darkening only (A) and (D) will result in +2 marks; and darkening (A) and (B) will result in –2 marks, as a wrong option is also darkened

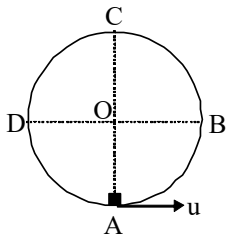
7. The system of two weights with masses m_1 and m_2 are connected with weightless spring as shown. The system is resting on the support S. The support S is quickly removed. Find the accelerations a_1 & a_2 of the weights m_1 & m_2 respectively immediately after the support S is removed from below given options. Use: $m_1 = m_2 = m$



(A) $a_2 = 2g$ (B) $a_1 = 0$ (C) $a_2 = g$ (D) $a_1 = g/2$

Space for Rough Work

8. ABCD is a smooth fixed vertical circular track. A small particle is projected tangentially from the lower most point A of the track with speed u = $\sqrt{4gr}$ where r is the radius of track. AC is the vertical diameter of track, then



(A) The particle leaves the track somewhere between B and C on the path of track

(B) The particle cannot hit the point C when it is moving in air

(C) The particle cannot cross the line AC when it moving in horizontal direction

(D) The particle cannot cross the line BD while moving vertically

9. Two narrow cylindrical pipes A and B are filled with hydrogen and helium gases respectively at same temperature. Pipe A is open at both ends and pipe B is open at one end and closed at the other end. If frequency of the second overtone of pipe A is equal to the frequency of the first overtone of the pipe B,

(A) ratio of the velocities of sound waves in the two pipes $\frac{v_A}{v_B} = \sqrt{2}$

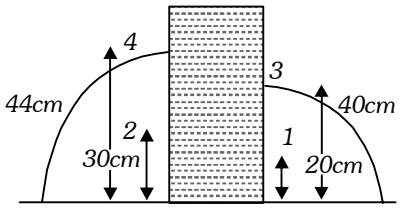
(B) $\frac{v_A}{v_B} = \sqrt{\frac{42}{25}}$

(C) ratio of their lengths $\frac{l_A}{l_B} = \sqrt{\frac{168}{25}}$

(D) $\frac{l_A}{l_B} = 2\sqrt{2}$

Space for Rough Work

10. Figure shows a cylindrical vessel of height 84 cm filled upto the brim. Cylinder is open from top. There are four holes in the vessel as shows in figure. The liquid falling at maximum horizontal distance from the vessel would come from:



(A) hole 1 (B) hole 2 (C) hole 3 (D) hole 4

11. Choose the **CORRECT** statements from the following.

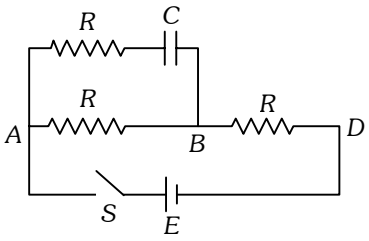
(A) When an electron is emitted from a nucleus, then its neutron/proton ratio is decreased.

(B) Number of protons in the nucleus is conserved during α-decay.

(C) In a hydrogen atom following Bohr model, the acceleration of electron in n = 2 orbit is less than that in n = 1 orbit.

(D) Potential energy of an electron in an atom decreases with increasing principal quantum number.

12. The switch is closed at t = 0 in the adjoining circuit. Which of the following is correct?



(A) The potential difference across BD at t = 0 is 2E/3

(B) The potential difference across AB at t = ∞ is E/2

(C) The potential difference across BD at t = 0 is E/2

(D) The potential difference across AB at t = ∞ is 2E/3

Space for Rough Work

13. Each of the following system begins moving upwards with a constant acceleration. Select these cases in which quantity will not change due to this upward acceleration :

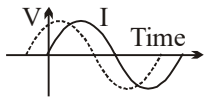
(A) time period of simple pendulum.

(B) fraction of floating body submerged in a liquid.

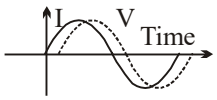
(C) time period of a spring block system.

(D) pressure on the base of a container containing liquid.

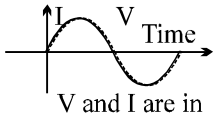
14. The diagram shows the variation of V and I in an AC circuit. The circuit only be a series RC or series RL or series LC or series RLC. Consider the four different combinations of V and I graphs. Pick the correct combination/combinations for each graph. Solid curves represent I and broken curves represent V.

(a) 

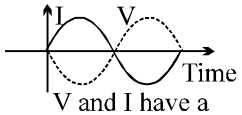
Phase difference between V & I is $\pi/4$

(b) 

Phase difference between V & I is $\pi/4$.

(c) 

V and I are in the same phase

(d) 

V and I have a phase difference of π

Take the angular frequency of the AC voltage source to be 100 rad/s (This is ω)

(I) : R = 1kΩ, L = 1H and C = 100μF

(II) : R = 1Ω and L = 10^{-2}H

(III) : R = 100Ω and C = $10^{+2}\mu$F

(IV) : L = 1H, C = 100μF and R = 1Ω

(A) a $\rightarrow$ II ; d $\rightarrow$ I, IV

(B) c $\rightarrow$ I ; d $\rightarrow$ none

(C) b $\rightarrow$ III ; c $\rightarrow$ IV

(D) a $\rightarrow$ II,III ; b $\rightarrow$ II,III

Space for Rough Work

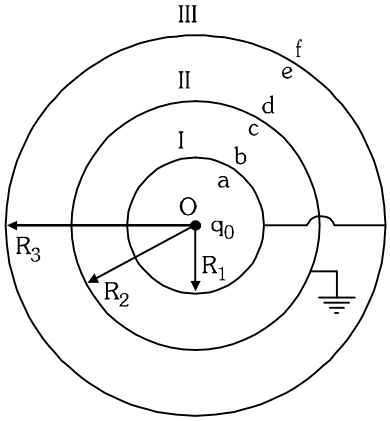
SECTION–I(iii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

Paragraph for Questions 15 and 16

Shells I, II, III are conducting spherical neutral shells of radius R, 2R and 3R respectively. Shell I and shell III are connected. A point charge q_0 is now kept at centre O. Now Shell II is also grounded. q_d represents charge on outer surface of shell-II. q_b represents charge on outer surface of shell-I.



15. Choose the correct option

(A) $q_d = -3q_b$.
(B) The total charge on shell II is zero.
(C) $q_d = -q_b$
(D) $q_d = +3q_b$

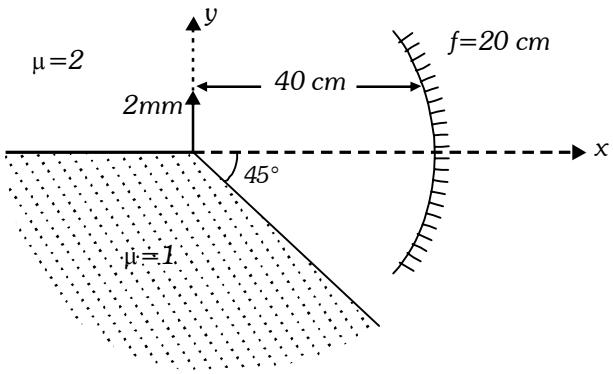
16. The charge on surface 'b' (outer surface of Shell I) is

(A) (2/11) q_0 (B) (3/11) q_0 (C) (1/11) q_0 (D) none of these

Space for Rough Work

Paragraph for Questions 17 and 18

Consider an object as the head of the shown arrow of height 2mm. Assume only paraxial rays to be incident. If the first reflection is to be taken on mirror then on interface to create an image at (x, y).



17. Find 2x + y :

(A) 2 mm (B) 2 cm (C) 4 mm (D) 4 cm

18. The created final image is -

(A) Real, enlarged then object (B) Real, same size as object

(C) Virtual, enlarged then object (D) Virtual, diminished wrt object

Space for Rough Work

PART-2 : CHEMISTRY

SECTION–I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks : –1 In all other cases

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1. The element gallium (Ga) has the second largest liquid range of any element, melting at 29.78°C and boiling at 2204°C at atmospheric pressure.

Assume that you construct a thermometer using, gallium as the fluid instead of mercury, and that you define the melting point of gallium as 0°G and the boiling point of gallium as 1000°G.

What is the melting point of sodium chloride (801°C) on the gallium scale ?

(A) 801 (B) 368 (C) 771.2 (D) 354.7

2. Compound X ($C_7H_{13}Br$) is a tertiary bromide. On treatment with potassium ethoxide in ethanol, X is converted into Y (C_7H_{12}). Ozonolysis of Y gives Z as the only product. Z can also be obtained by the oxidation of heptan-2, 6-diol.

Which statement about Z is incorrect.

(A) Z can also be obtained by oxidation of Y with conc. hot $KMnO_4$ solution.

(B) Z gives negative 2, 4-DNP test.

(C) Z gives positive iodoform test.

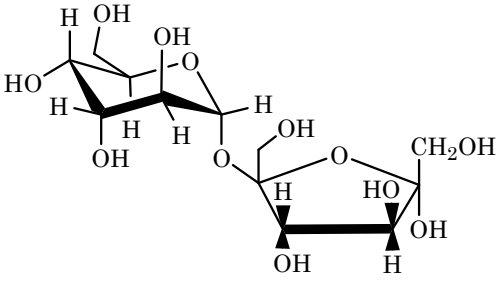
(D) Z can be reduced to a diol with $NaBH_4$

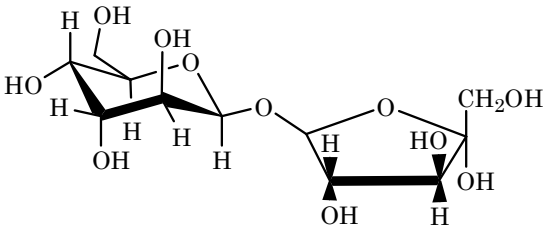
Space for Rough Work

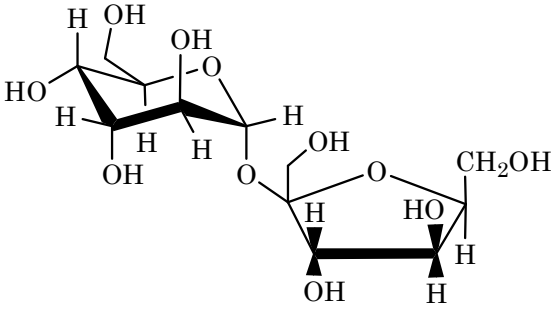
3. Which of the following disaccharides fulfills the two requirements stated below ?

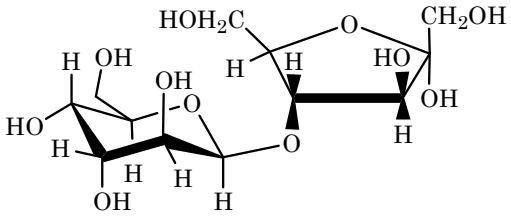
(i) α–(1,2) glycosidic linkage

(ii) Not a reducing sugar

(A) 

(B) 

(C) 

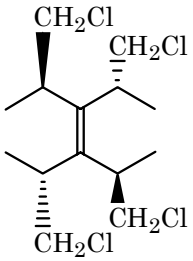
(D) 

Space for Rough Work

4. When an electron is removed from every ion of 1 mol of H^- ions, 72 kJ of energy is required. In another experiment, when electrons are removed from $\frac{1}{2}$ mole of H (one electron each) and given them to another $\frac{1}{2}$ mole of H (one electron each), 619.5 kJ of energy is found to be required. The ionisation energy (in kJ/mol) of hydrogen is :

(A) 691.5 (B) –691.5 (C) 1311 (D) –1311

5. True statement(s) about this compound is/are:



(A) It is E isomer (B) It is Z isomer

(C) It is optically active (D) It has center of symmetry

6. A molecule M associates in a given solvent according to the equation $M \rightleftharpoons (M)_n$. For a certain concentration of M, the van't Hoff factor was found to be 0.9 and the fraction of associated molecules was 0.2. The value of n is

(A) 28/18 (B) 18/28 (C) 5 (D) 1/5

Space for Rough Work

SECTION–I(ii) : (Maximum Marks : 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
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Partial Marks : +1 For darkening a bubble corresponding **to each correct option**, Provided NO incorrect option is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks : –2 In all other cases.

- for example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in +4 marks; darkening only (A) and (D) will result in +2 marks; and darkening (A) and (B) will result in –2 marks, as a wrong option is also darkened

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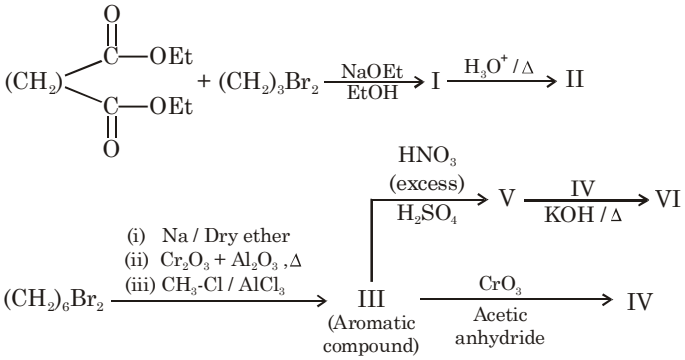
7. When excess of silver nitrate is reacted with I_2 in aqueous solution then which of the following compounds are formed ?

(A) $AgIO_3$ (B) AgI (C) HIO_3 (D) HNO_3

8. Which of the following is/are used as flux to remove impurities in metallurgical processes

(A) Silica (B) Limestone (C) NaCl (D) Coke

9.



(A) II $\xrightarrow[(ii)\,Br_2\\(iii)\,Mg\\(iv)\,H_3O^+]{(i)\,Ag_2O}$ product is having boiling point more than cyclopropane.

(B) VI gives optically active compound by reaction with D_2(1 eq.) / Pt.

(C) II is resolvable

(D) Index of Hydrogen Defficiency of VI is 10

Space for Rough Work

10. A Daniel Cell is made at 25°C by connecting Zn / Zn^{+2} (0.1 M, 1 litre) & Cu / Cu^{+2} (0.9 M, 1 litre) electrode. The cell is discharged till its emf reaches 1.10 volt, then it is charged (reversing the discharge process) by passing 0.6 F charge.

[**Given** : $E^{o}_{Zn/Zn^{+2}} = 0.76$ V, $E^{o}_{Cu/Cu^{+2}} = -0.34$ V, $\frac{2.303RT}{F} = 0.06$, log2 = 0.30]

Select the correct option(s).

(A) Concentration of Zn^{+2} becomes 1 M when EMF reach 1.10 volt.

(B) Concentration of Cu^{+2} becomes 0.5 M when EMF reach 1.10 volt.

(C) EMF of the cell just after charging is 1.118 volt

(D) After charging concentration of Cu^{+2} becomes 0.8 M.

11. Select incorrect statements :

(A) Pre–exponential factor for zero order reaction is a unitless quantity.

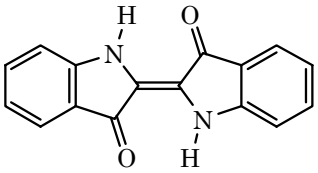
(B) If $t_{1/4}$ = 30 sec then $t_{1/2}$ = 60 sec for first order reaction

(C) If $t_{1/3}$ = 30 sec then $t_{2/3}$ = 90 sec for sec–order reaction

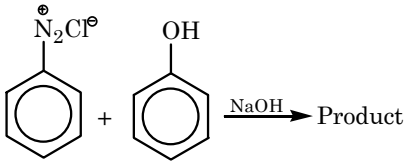
(D) If $t_{1/5}$ = 30 sec then $t_{3/5}$ = 90 sec for zero order reaction.

Space for Rough Work

12. **Correct** statements about indigotin is/are



(A) It shows geometrical isomerism
(B) Trans is more stable than cis
(C) Cis is more stable than trans
(D) It is a planar compound

13. 

Which of the following statements is/are correct about the above reaction ?

(A) Product shows geometrical isomerism.

(B) Product shows colour due to extended conjugation.

(C) Electophile attack at para position due to large size.

(D) Electron withdrawing group in phenol increases rate of reaction.

14. Which of the following are square planar complexes ?

(A) $[Ni(CO)_4]$ (B) $[PtCl_2(NH_3)_2]$ (C) $[Cu(NH_3)_4]^{2+}$ (D) $[FeCl_4]^-$

Space for Rough Work

SECTION–I(iii) : (Maximum Marks : 12)

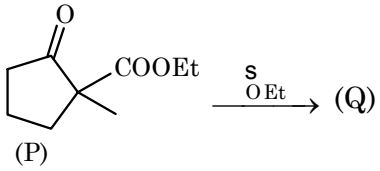
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Paragraph for Questions 15 and 16

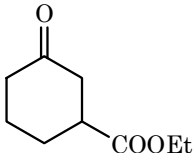
Ester having α–'H' gives β– keto ester in presence of alkoxide ion. This reaction is known as Claisen condensations reaction.

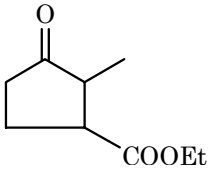
15.

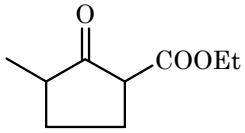


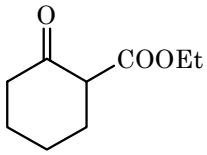
(P)

Possible structure of **"Q"** is

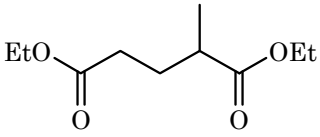
(A)


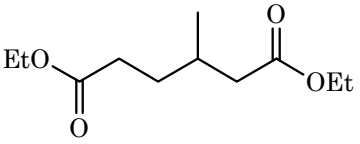
(B)


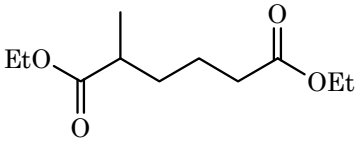
(C)


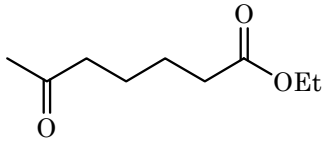
(D)


16. Which of the following compound is obtained during above interconversion.

(A)


(B)


(C)


(D)


Space for Rough Work

Paragraph for Questions 17 and 18

An aqueous solution of salt **(P)** gives white precipitate with Barium chloride solution which is insoluble in dilute HCl & dilute HNO_3. When ammonia (aq) is added to the aqueous solution of salt **(P)** blue precipitate is obtained which dissolves in excess ammonia producing deep blue colouration due to formation complex **(Q)**. With KCN, **(P)** gives first a yellow precipitate which quickly decomposes to white precipitate along with liberating highly poisonous gas **(R)**. **(R)** on hydrogenation produces **(S)** which gives yellowish orange coloured complex **(T)** with $CoCl_3$.

17. Which of the following options is **CORRECT** with respect to P, Q, R, S and T.

(A) 'P' is Ag_2SO_4, 'Q' is $[Ag(NH_3)_2]_2SO_4$ (B) 'S' is a bidentate ligand & 'R' is cynogen

(C) 'R' is CH_2N_2 & 'P' is $CuSO_4$ (D) None of the above options is correct

18. Which of the following options regarding complex Q & T is **NOT CORRECT**.

(A) Q is paramagnetic where as T is diamagnetic.

(B) Q has a square planar geometry where as T has octahedral geometry.

(C) When dissolved in water, Van't Hoff coefficient for Q is 2 & that of T is 4 assuming 100% dissociation.

(D) Only one of the above statements is correct.

Space for Rough Work

PART-3 : MATHEMATICS

SECTION–I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

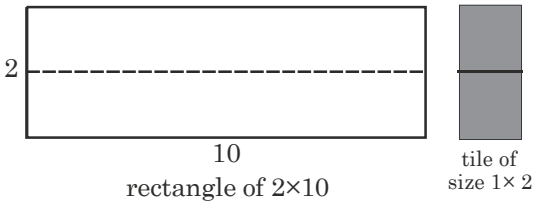
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases

---

1. Let a, b, c, d are roots of polynomial $x^4 + 5x^3 - 4x^2 + 6 = 0$ then the value of $\frac{1}{a^2-5a+6}+\frac{1}{b^2-5b+6}+\frac{1}{c^2-5c+6}+\frac{1}{d^2-5d+6}$ is :

(A) $\frac{-76}{46}$ (B) $\frac{-219}{186}$ (C) 0 (D) $\frac{677}{1426}$

2. A 2 × 10 rectangle needs to be tiled with tiles of size 1 × 2 (domino) as shown in figure. In how many ways can this be done?



(A) 34 (B) 55 (C) 89 (D) None of these

3. Number of pairs (x, y) of real numbers that satisfy the equation

$$\left(\sin^2 x+\frac{1}{\sin^2 x}\right)^2+\left(\cos^2 x+\frac{1}{\cos^2 x}\right)^2=12+\frac{1}{2}\sin y$$

(A) 0 (B) 1 (C) 2 (D) Infinite

Space for Rough Work

4. Let $z_0, z_1, z_2, \ldots, z_6$ be distinct complex numbers satisfying $z^7 - 7 = 0$. If p is the product of all sums of z_i taken pairwise i.e. $p = (z_0 + z_1).(z_0 + z_2)\ldots\ldots(z_5 + z_6)$, then p^2 is

(A) 7^3 (B) 7^6 (C) $2^{21} \cdot 7^3$ (D) None of these

5. $\dfrac{\sum_{n=1}^{99} \sqrt{10 + \sqrt{n}}}{\sum_{n=1}^{99} \sqrt{10 - \sqrt{n}}}$ equals

(A) $\sqrt{2}$ (B) $\sqrt{2} - 1$ (C) $\sqrt{2} + 1$ (D) None of these

6. Let A, B, C, D be randomly selected subsets of X = {1, 2, 3, ..., *n*} such that $A \cap C = \phi, B \cap D = \phi$. The probability of $A \cap B \neq \phi, B \cap C \neq \phi, C \cap D \neq \phi$ and $D \cap A \neq \phi$, is

(A) $(9^n - 8^n + 7^n - 6^n + 5^n)/9^n$ (B) $(9^n - 4.8^n + 6.7^n - 4.6^n + 5^n)/9^n$

(C) $(7^n - 2.6^n + 5^n)/7^n$ (D) None of these

Space for Rough Work

SECTION–I(ii) : (Maximum Marks : 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened.

Partial Marks : +1 For darkening a bubble corresponding **to each correct option**, Provided NO incorrect option is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks : –2 In all other cases.

- for example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in +4 marks; darkening only (A) and (D) will result in +2 marks; and darkening (A) and (B) will result in –2 marks, as a wrong option is also darkened

---

7. Let matrix $A=\begin{bmatrix} x & y & -z \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$, where $x,y,z \in N$ If $|\text{adj}(\text{adj}(\text{adj}(\text{adj}A)))| = 4^8 \cdot 5^{16}$, then the number of such matrix is p and value of $\int\limits_{-\pi/2}^{\pi/2} 2^{\sin x}dx + \int\limits_{5/2}^{4} \sin^{-1}(\log_2(x-2))dx$ is equal to $\frac{q\pi}{4}$.

If $\vec{a}=\hat{i}+\hat{j}+2\hat{k}$, $b=\hat{i}+2\hat{j}+2\hat{k}$ and $|\vec{c}|=1$ such that $[\vec{a}\times\vec{b}\ \ \vec{b}\times\vec{c}\ \ \vec{c}\times\vec{a}]$ has maximum value then

(A) the value of $\left|(\vec{a}\times\vec{b})\times\vec{c}\right|^2$ is 0 (B) p = 36

(C) q = 5 (D) p – q = 20

8. For differential equation of $y=c_1\left(e^{\sqrt{x}+c_2}+e^{-\sqrt{x}+c_2}\right)$:

(A) Order = 1 (B) Order = 2 (C) Degree = 1 (D) Degree = 2

Space for Rough Work

9. Identify which of the following statement(s) is(are) correct.

(A) $\lim\limits_{x\to 0}\dfrac{x^{6000}-(\sin x)^{6000}}{x^2\cdot(\sin x)^{6000}}$ is equal to 1000

(B) $\lim\limits_{x\to 0}\dfrac{x^{6000}-(\sin x)^{6000}}{x^2\cdot(\sin x)^{6000}}$ is equal to 3000

(C) $\lim\limits_{x\to 0}\dfrac{1-\cos^5 x\cos^3 2x\cos^3 3x}{x^2}$ is equal to 11

(D) $\lim\limits_{x\to 0}\dfrac{1-\cos^5 x\ \ \cos^3 2x\ \ \cos^3 3x}{x^2}$ is equal to 22

10. Find polynomials $f(x), g(x)$ and $h(x)$ If they exist such that for all x,

$$|f(x)|-|g(x)|+h(x)=\begin{cases}-1 & \text{if } x<-1\\ 3x+2 & \text{if } -1\le x\le 0\\ -2x+2 & \text{if } x>0\end{cases}$$

(A) $f(x)=3x+3$ (B) $g(x)=\dfrac{5x}{2}$ (C) $h(x)=-x+\dfrac{1}{2}$ (D) None of these

11. If equation $16x^4-mx^3+(2m+17)x^2-mx+16=0$ has four distinct roots forming a geometric progression, and one of the root is $\dfrac{1}{2}$, then :

(A) common ratio of G.P. is 4 (B) m = 170

(C) sum of the roots = $\dfrac{85}{8}$ (D) Product of root = 3

Space for Rough Work

12. A leaf is torn from a paperback novel. The sum of the numbers on the remaining printed pages is 15,000 :

(A) sum of page numbers of the torn leaf is 51

(B) number of printed pages in the book is 174

(C) number of printed pages in the book is 173

(D) sum of page numbers of the torn leaf is 55

13. Pankaj and Prateek are Algebra genius. In a game Pankaj is going to pick three non-zero real numbers and Prateek is going to arrange the three numbers as the coefficients of a quadratic equation $ax^2 + bx + c = 0$. Pankaj wins the game if and only if the resulting equation has two distinct rational solutions else Prateek will win.

(A) Prateek has 100% chance of winning

(B) Pankaj has 100% chance of winning

(C) both have 50% chance of winning

(D) 1 may be a root for winning strategy

14. For the equation $2(2^x - 1)x^2 + (2^{x^2} - 2)x = 2^{x+1} - 2$ for real numbers x

(A) Number of real solutions = 4

(B) Number of real solutions = 3

(C) Sum of real solutions = 0

(D) Sum of real solutions = 2

Space for Rough Work

SECTION–I(iii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

Paragraph for Questions 15 and 16

The gamma function $y = \Gamma(x)$ is a continuous smooth function in practice that connects the points (x, y) given by $y = (x-1)!$ at the positive integer values for x. It is uniquely defined by $\Gamma(x) = \int_0^\infty t^{x-1}e^{-t}dt$, for every real x except non positive integers together with $\Gamma(1) = 1$ and $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$. *Another important function is beta function* $B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}\,dt = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$ *for all* $x,y \in \mathbb{R}^+$. *Now answer the following questions :*

15. $\int_0^{\pi/2} \sin^{10}x.\cos^{12}x\,dx$

(A) $\frac{3^4.5.7^3}{2^{16}}\pi$ (B) $\frac{3^4.5.7^4}{2^{16}}\pi$ (C) $\frac{3^4.5^2.7^2}{2^{16}}\pi$ (D) None of these

16. $\int_0^\infty t^b e^{-at}\,dt =$ $(b \in N)$

(A) $\frac{\Gamma(b+1)}{a^{b+1}}$ (B) $\frac{\Gamma(a)\Gamma(b)}{ab}$ (C) $\frac{\Gamma(b+1)\Gamma(a+1)}{a^{b+1}}$ (D) $\frac{\Gamma(a+1)}{a^{b+1}}$

Space for Rough Work

Paragraph for Questions 17 and 18

For all $x \geq 1$, let $f(x)$ be the function defined as $f(x) = \begin{cases} [x] \cdot \left| x - [x] - \dfrac{1}{2[x]} \right|, & \text{if } x < [x] + \dfrac{1}{[x]} \\ f\left(x - \dfrac{1}{[x]} \right) & \text{otherwise} \end{cases}$

(where [.] denote greatest integer function)

17. The function $f(x)$ must satisfies the condition

(A) $f(4) = f\left(4\frac{1}{8}\right)$ (B) $f\left(4\frac{1}{4}\right) = f\left(4\frac{1}{8}\right)$

(C) $f\left(4\frac{1}{3}\right) = f(4)$ (D) $f(4) = f\left(4\frac{1}{4}\right)$

18. The number of points at which f(x) is not differentiable in the interval $x \in (4, 5)$ is

(A) 8 (B) 7 (C) 4 (D) 2

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.
17. Each part has one section as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	6	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	18
I(ii)	One or more correct option(s)	8	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–2 In all other cases	32
I(iii)	Paragraph Based (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 In all other cases	—	12

NAME OF THE CANDIDATE .....

FORM NO. .....

I have read all the instructions and shall abide by them.	I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.
_____	_____
Signature of the Candidate	Signature of the Invigilator

Paper Code : 1001CJA102117076

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2021 - 2022)

JEE (Main + Advanced) : ENTHUSIAST COURSE
SCORE-I

Test Type : FULL SYLLABUS Test Pattern : JEE-Advanced

TEST DATE : 16 - 06 - 2021

PAPER-1

PART-1 : PHYSICS SOLUTION

SECTION-I

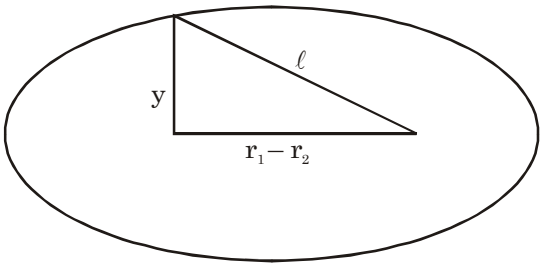
1. Ans. (B)

Sol. $\frac{V_1}{V_2}=\frac{L_1i_1+Mi_2}{L_2i_2+Mi_1}$

$=\frac{L_1i_1+\sqrt{L_1L_2}i_2}{L_2i_2+\sqrt{L_1L_2}i_1}=\sqrt{\frac{L_1}{L_2}}$

2. Ans. (C)

Sol.



$\ell^2=y^2+(r_1-r_2)^2$ ...(i)

$y+\ell=r_1+r_2$ ...(ii)

$\Rightarrow y=\frac{r_1r_2}{r_1+r_2}$

3. Ans. (D)

Sol. Apply work energy

$\frac{mgr}{2}=\frac{1}{2}(2mr^2)\omega^2$

$\omega=\sqrt{\frac{g}{2r}}$

$v_{cm}=\frac{\omega r}{2}=\sqrt{\frac{gr}{8}}$

4. Ans. (C)

Sol. $P\propto V$ (given)

$PV=nRT$ gives

$V^2\propto T$

$\Rightarrow \frac{V_1^2}{T_1}=\frac{V_2^2}{T_2}$

$T_2=\frac{V_2^2T_1}{V_1^2}=\frac{(2v)^2T_1}{V^2}=4T_1$

$\therefore \Delta T=T_2-T_1=3T_1=3\times300=900\ K$

and $\Delta U=nC_v\Delta T=1\times\frac{3}{2}R\times900=1350\ R$

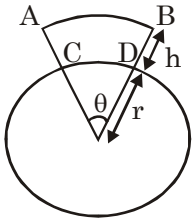
5. *Ans. (B)*

Sol. *use fourier's equation* $-kA\frac{dT}{dx}=\frac{dQ}{dt}\Big|_{steady}$

6. Ans. (B,C,D)

7. Ans. (B,D)

Sol.



Comparing optical path of AB & CD

$(r+h)\frac{n_0}{1+\frac{h}{2R}}=rn_0$

$1+\frac{h}{r}=1+\frac{h}{2R}$

$r=2R$

8. Ans. (A,C)

9. Ans. (A,C)

Sol. $k\left(x+\frac{\ell}{2}\right)^2\theta=\left(\frac{m\ell^2}{12}+mx^2\right)\alpha$

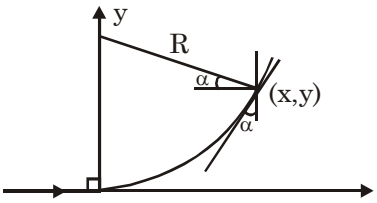
$$\alpha=\frac{k\left(x+\frac{\ell}{2}\right)^2}{\frac{m\ell^2}{12}+mx^2}\theta$$

$$\omega^2=\frac{\left(x+\frac{\ell}{2}\right)^2 k}{\frac{m\ell^2}{12}+mx^2}$$

$$\frac{d\omega^2}{d\theta}=0 \Rightarrow x=\frac{\ell}{6},\ \omega=\sqrt{\frac{4k}{m}}$$

10. Ans. (B,C)

Sol.



(I) $\sin 90 = (n)\frac{R-y}{R}$

$$n=\frac{R}{R-y}$$

$$\frac{dn}{dy}=\frac{R}{(R-y)^2}$$

(II) for n = 2.5

$$y=\frac{3R}{5}$$

11. Ans. (A,B)

Sol. Use $I = 1/2\ \rho\omega^2A^2\ v$

$P = 1/2\ \mu\omega^2A^2\ v$

$ED = 1/2\ \rho\omega^2A^2$

12. Ans. (A,C,D)

Sol. $P_{heater} = P_{given\ to\ block} + P_{Loss\ to\ surroundings}$

Initial $P_{Loss} = 0$

$\therefore \quad 500 = ms\frac{dT}{dt} + 0$

$500 = C\ 2.5$

$\therefore \quad C = 200\ J/°C$

At 50°, power loss to surroundings = 100W

$$0 = ms\frac{dT}{dt} + 100$$

$$\frac{dT}{dt} = -\frac{100}{200} = -0.5\ s/sec$$

Given at 50°C : Newton's Law of cooling

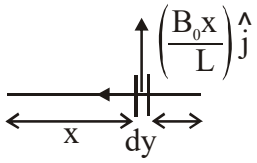
Power Loss = 100 = k(50 – 10)

$$\Rightarrow \quad k = \frac{10}{4} = \frac{5}{2}$$

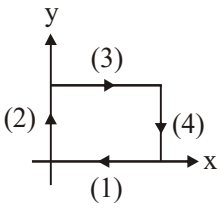
$$\therefore \text{ At } 30°C\ P_{Loss} = k(30-10) = \frac{5}{2}\times 20 = 50\ W$$

13. Ans. (A,B,C,D)

Sol. Wire segment (1)



$$(d\vec{f}) = \int_0^L i\,dx\left(\frac{B_0x}{2}\right) = \frac{iB_0L}{2}$$



$$d\vec{f} = \frac{-iB_0L}{2}\hat{k}$$

Similary for $\vec{f}_2 = -\frac{-iB_0L}{2}\hat{k}$

$$\vec{f}_3 = \frac{iB_0L}{2}\hat{k}$$

$$\vec{f}_4 = \frac{iB_0L}{2}\hat{k}$$

$$\vec{f}_n = 0$$

If coil is constrained to rotate about

$$\text{y-axis } |\bar{\tau}| = 1\left(\frac{B_0L}{2}\right)L = \frac{iAB_0}{2}$$

$\Rightarrow$ Similarly for torque about torque has same magnitude

SECTION - IV

1. Ans. 6

Sol. $2T\theta = \mu(R2\theta)\omega^2R$

$T = \mu\omega^2R^2$

$T = \frac{M\omega^2R}{2a}$ take log

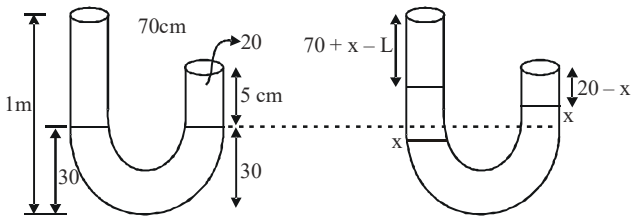
$\log T = \log\frac{MR}{2\pi} + 2\log\omega$

differentiate

$\frac{\Delta T}{T} = 2\frac{\Delta\omega}{\omega}$ % change = 6%

2. Ans. 6

Sol. $2x \times \frac{\rho}{2}g = L \times \rho g$



$\Rightarrow x = L$

frequency same $\Rightarrow \lambda$ = same

$20 - x = \frac{\lambda}{4}$ $\qquad 70 = \frac{5\lambda}{4}$

$70 = 100 - 5x$

$5x = 30$; $x = 6$ cm

3. Ans. 3

Sol. $175\left(\frac{c}{c-v} - \frac{c}{c+v}\right) = 3$

$175 \times 2vc = 3c^2 - 3v^2$

assuming $v << c$

$350\ vc = 3c^2$

$\therefore$ $v = 3$m/s

4. Ans. 5

5. Ans. 5

Sol. Process A $\rightarrow$ B

$W_{AB} = \int P\,dv = \int\frac{3}{2}T^{1/2}dv$

$= \int\frac{3}{2}T^{1/2} \times \frac{1}{3}RT^{-1/2}dT$

On solving, $W_{AB} = 50\ R = 50 \times 8.3 = 415$ J

Process B $\rightarrow$ C

$U = \frac{1}{2}V^{1/2}$

$\frac{3}{2}RT = \frac{1}{2}V^{1/2}$

$\Rightarrow$ $3PV^{1/2} = 1$

$\therefore$ $P = \frac{1}{3\sqrt{V}}$

Now $W_{BC} = \int P\,dv = \int_{100}^{1600}\frac{1}{3\sqrt{V}}dv$

$= \frac{2}{3}\sqrt{V} = \frac{2}{3}[40 - 10] = \frac{2}{3} \times 30 = 20$ J

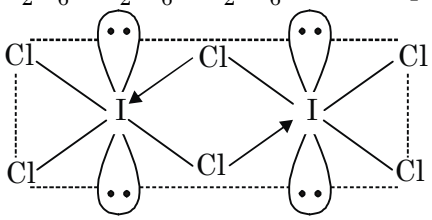
Total W = 415 + 20 = 435]

PART–2 : CHEMISTRY SOLUTION

SECTION-I

1. Ans.(D)

Sol. B_2H_6, Al_2Cl_6, Al_2Br_6 are non-planar with sp^3 centre and I_2Cl_6 is planar with sp^3d^2 centre.



2. Ans.(C)

Sol. $P_{\text{gas C}}$ at 300 K $= \dfrac{300}{400} \times 600 = 450\,\text{mm Hg}$

Remaining P = 485 – 450 = 35 mm Hg gas 'A' in flask - I.

Let 'x' is number of moles of gas 'A' in flask - I

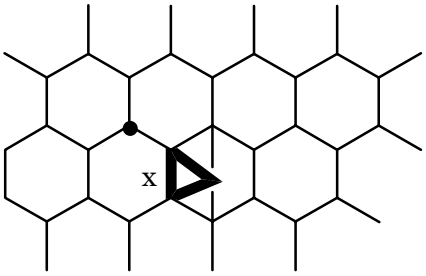
$\Rightarrow$ no. of mole of gas A in flask- II $= \dfrac{3}{4}x$

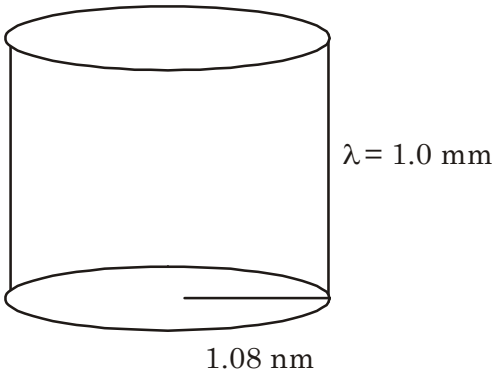
mole fraction of total mole remain in 'flask - I' $= \dfrac{4}{7}$

Pressure in 'flask - I' $= \dfrac{4}{7} \times 35 = 20\,\text{mm Hg}$

3. Ans. (D)

Sol.





$$\text{Number of hexagon} = 2 \times \frac{1}{2} \frac{2 \times 3.14 \times 1.08 \times 10^{-9} \times 1 \times 10^{-3}}{6 \times \frac{\sqrt{3}}{4} \times \left(141.5 \times 10^{-12}\right)^2} \times 10^{12} = 1.3039 \times 10^{8}$$

4. Ans.(D)

Sol. Theory based.

5. Ans. (B)

Sol. AgCl is not soluble in H_2O because lattice energy is more than hydration energy.

$$AgCl + 2KCN \longrightarrow \underset{\text{soluble}}{K[Ag(CN)_2]} + KCl$$

$$Zn + 10HNO_3(v\,dil) \longrightarrow 4Zn(NO_3)_2 + NH_4NO_3 + 3H_2O$$

$$2CuSO_4 + 4KI \longrightarrow Cu_2I_2 + I_2 + 2K_2SO_4$$

$$\underset{\text{blue}}{CuSO_4.5H_2O} \xrightleftharpoons{\text{exposes}} \underset{\text{pale blue}}{CuSO_4.3H_2O} \xrightleftharpoons{100^\circ C} \underset{\text{bluish white}}{CuSO_4.H_2O}$$

$$\xrightleftharpoons{230^\circ C} \underset{\text{white}}{CuSO_4} \xrightarrow[\Delta]{720^\circ C} CuO + SO_2 + \frac{1}{2}O_2$$

6. Ans.(A,B,C)

Sol. (A) $2Hg + O_3 \longrightarrow Hg_2O + O_2$

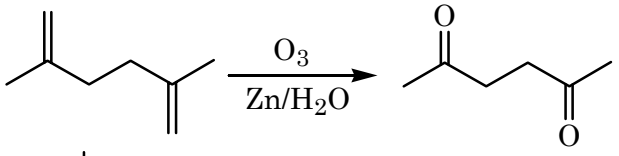
(B) C_3O_2 (carbon suboxide) O=C=C=C=O

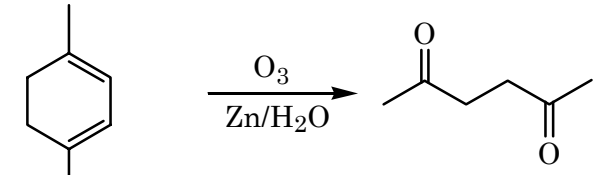
(C) O^{2-} and O_2^{2-} are diamagnetic but O_2^- and O_3^- are paramagnetic

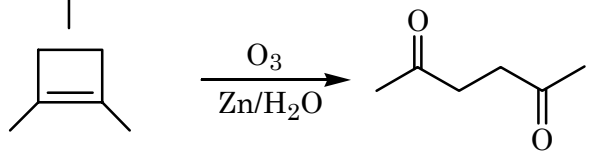
(D) Mn_2O_7 is acidic but MnO is basic not amphoteric

7. Ans.(A,B,C)

Sol.

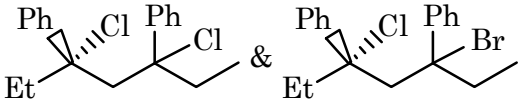






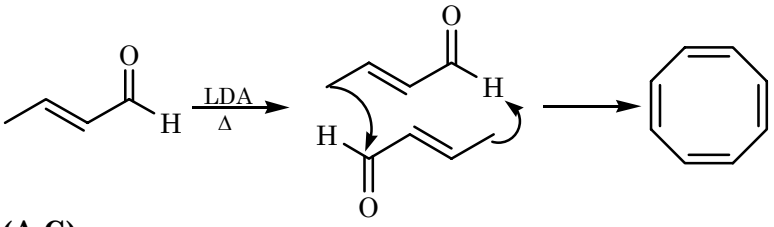
8. Ans. (C,D)

Sol. Products are



9. Ans. (B,C)

Sol. In the absence of α–H on saturated carbon, γ–H becomes acidic. The acidic γ–H would be abstracted by LDA to give carbanion, which condense with another molecular to give cycloctatetraene. The high temperature reaction condition helps in the dehydration of aldol product.



10. Ans. (A,C)

Sol. Theory based.

11. Ans.(A,B,D)

Sol. (A) Temperature T_0, represent tripple point of solvent where Solid $\rightleftharpoons$ Liquid $\rightleftharpoons$ vapour are in equilibrium hence vapour pressure of solid arid vapour pressure of liquid will be same.

(B) Figure showing $P_0 > P_1 > P_2$

(C) Line corresponding to P_0 and P_2 vapour pressure $\text{Solid} \rightleftharpoons \text{vapour}$

$$\text{Hence ln}\ \frac{P_2}{P_0} = \frac{\Delta H_{sub}}{R}\left[\frac{1}{T_0} - \frac{1}{T_1}\right]$$

(D) Line corresponding to P_1 and P_2 vapour pressure $\text{Liquid} \rightleftharpoons \text{vapour}$

$$\text{Hence ln}\ \frac{P_2}{P_1} = \frac{\Delta H_{vap}}{R}\left[\frac{1}{T_0} - \frac{1}{T_1}\right]$$

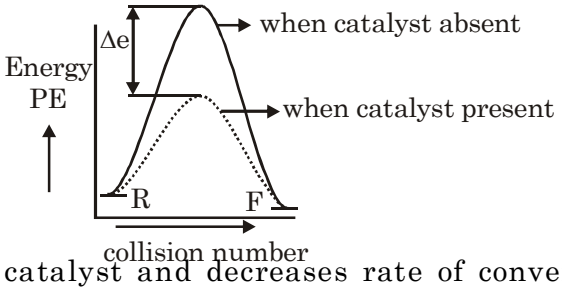
12. Ans. (A,B,C,D)

Sol. (A) is correct

$$k_p = Ae^{-E_p/RT}$$

$$k_a = Ae^{-E_a/RT}$$

$$\Rightarrow \quad \frac{k_p}{k_a} = e^{\Delta E/RT}$$

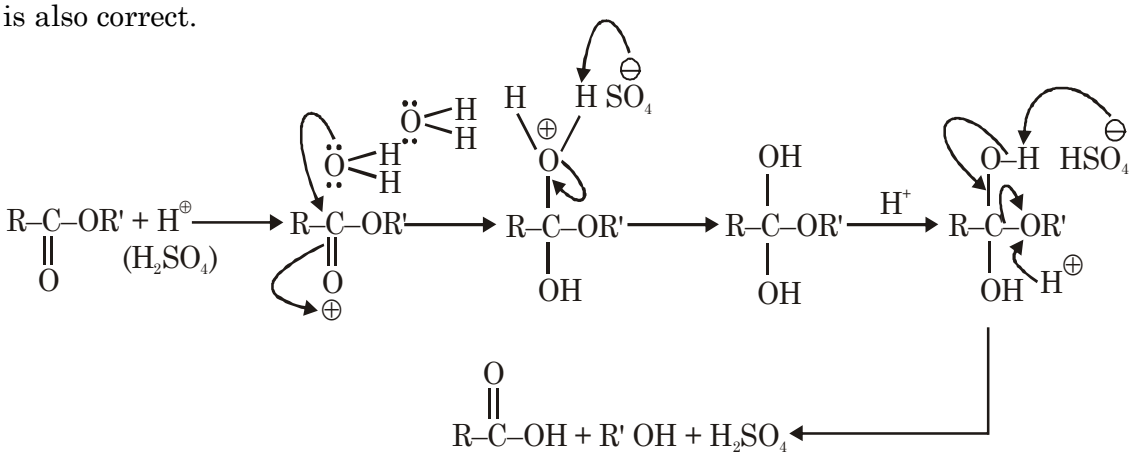


(B) is correct as alcohol is a negative catalyst and decreases rate of conversion of

$Na_2SO_3 \longrightarrow Na_2SO_4$

(C) is also correct as tetraethyl lead is an anti–knocking agent.

(D) is also correct.



13. Ans. (A,C,D)

Sol. In PF_3 , it accepts back bonding M → L into d-orbital of Phosphorus atom.

SECTION - IV

1. Ans.(3)

Sol. $q = q_{AB} + q_{BC} + q_{CD} + q_{DA}$

$$= nRT_B.\ell n\frac{2V_0}{V_0} + n.C_{v,m}.(T_C - T_B) + nRT_c\ell n\frac{V_0}{2V_0} + n.C_{v,m}(T_A - T_D)$$

$$= nR\ell n2(T_B - T_C) = 10.7 \times 2 \times 0.7 \times 200 \simeq 3\,\text{k cal}$$

2. Ans. (4)

Sol. $5Fe^{2+} + MnO_4^- + 8H^+ \longrightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$

$$\Delta G^o = -5 \times F \times E^o_{Cell}$$

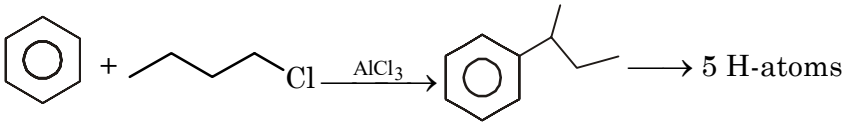
$$= -5 \times F \times 0.8 \text{ Joule}$$

$$\Delta G^{o} = \Delta G^{\circ}_{f\,Mn^{2+}} + 5\Delta G^{\circ}_{f\,Fe^{3+}} + 4\Delta G^{\circ}_{f\,H_2O(\ell)} - 5\Delta G^{\circ}_{f\,Fe^{2+}} - \Delta G^{\circ}_{f\,MnO_4^-} - 8\Delta G^{\circ}_{f\,H^+}$$

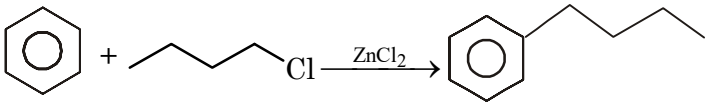
$$-5 \times 96500 \times 0.8 \times 10^{-3} = (-229) + 5(-10.2) + 4(-230) - 5(-84) - \Delta G^{\circ}_{fMnO_4^-}$$

$$\Delta G^{\circ}_{fMnO_4^-} = -780 + (5 \times 96500 \times 0.788 \times 10^{-3}) \approx 400.$$

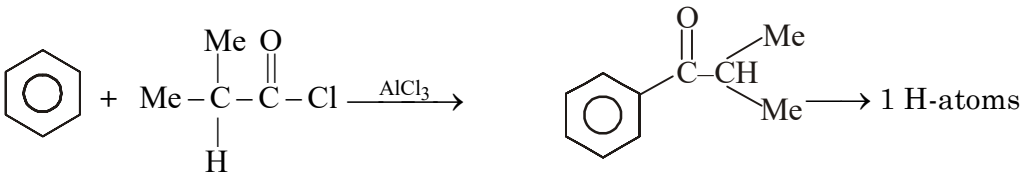
3. Ans. (8)

Sol. (a) 

→ 5 H-atoms

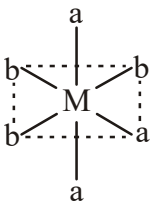
(b) 

→ 2 H-atoms

(d) 

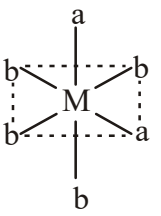
→ 1 H-atoms

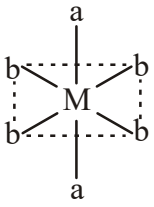
4. Ans. (6)

Sol. (a) 

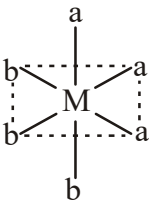
on replacing only one 'a' by 'b' possible configurations of the product are :

Meridional



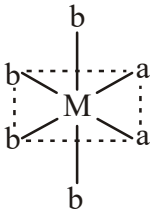
and


Cis Trans

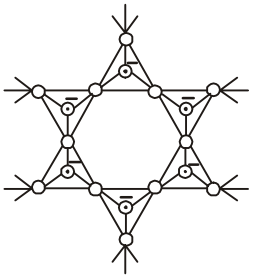


on replacing only one 'a' by 'b' possible configurations of the product

Facial

are :


Cis (only)

(b)  $\Rightarrow$ 2-D or sheet Silicate

Number of corner shared per Tetrahedron z = 3

x = 2

y = 1

z = 3

Total 2 + 1 + 3 = 6 Ans.

5. Ans.(7)

Sol.

$$A \xrightarrow{K=10^{-3}\,min^{-1}} 2B + C$$

t = 0	4	0	3
t = t	4–x	2x	3 + x

(4 – x) 40 + 2x (10) + (3 + x) (–30) = 0

x = 7/5

$$10^{-3} \times t = \ell n \frac{4}{4 - \frac{7}{5}}$$

t = 420/60 = 7

PART-3 : MATHEMATICS SOLUTION

SECTION-I

1. Ans. (D)

$$f(x) = \begin{cases} 4x+1; & -\infty < x \le \frac{1}{3} \\ 2+x; & \frac{1}{3} < x \le \frac{2}{3} \\ 4-2x; & \frac{2}{3} < x < \infty \end{cases}$$

$\therefore$ Clearly $f_{max}\left(x = \frac{2}{3}\right) = \frac{8}{3}$.

2. Ans. (A)

It is easy to see that x = 0 is a solution. Since the right hand side is a decreasing function of x and the left hand side is an increasing function of x, there is at most one solution. Thus x = 0 is the only solution to the equation.

3. Ans. (B)

Since $a_{n+2} = 1 - \frac{1}{a_{n+1}} = 1 - \frac{a_n}{a_n - 1} = \frac{-1}{a_n - 1}$, it follows that

$$a_{n+3} = 1 - \frac{1}{a_{n+2}} = 1 + (a_n - 1) = a_n,$$

so $\{a_n\}$ is a periodic sequence with a period 3.

Since $a_1 = 2, a_2 = \frac{1}{2}, a_3 = -1$, so $P_3 = -1$ and

$P_{2017} = (P_3)^{672} \cdot P_1 = 2$

The answer is **(B)**

4. Ans. (C)

$a^2 - a - 3 = 0$ yields $a^2 - a = 3$

On the other hand,

$a^3 + 1 = (a+1)(a^2 - a + 1)$,

$a^5 - a^4 - a^3 + a^2 = a^2(a^3 - a^2 - a + 1)$

$= a^2[(a^2(a-1) - (a-1)]$

$= a^2(a-1)(a^2-1) = a^2(a+1)(a-1)^2$

$= (a+1)(a^2 - a)^2$

it follows that

$$\frac{a^3+1}{a^5-a^4-a^3+a^2}=\frac{a^2-a+1}{(a^2-a)^2}=\frac{4}{3^2}=\frac{4}{9}$$

5. Ans. (D)

$$\left(\sqrt{2}p\hat{i}+q\hat{j}+r\hat{k}\right).\left(2\sqrt{2}\hat{i}+3\hat{j}+\hat{k}\right)$$

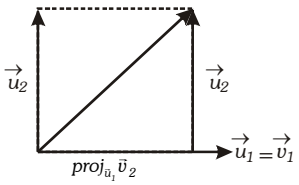
$$=\sqrt{2p^2+q^2+r^2}\sqrt{18}\cos\theta$$

$$\Rightarrow \quad \left(\sqrt{2p^2+q^2+r^2}\right)_{min}=\frac{7}{3\sqrt{2}}$$

when $\cos\theta = 1$

$$\therefore \quad (2p^2+q^2+r^2)_{min}=\frac{49}{18}$$

6. Ans. (A,C,D)



$\vec{u}_2=\vec{v}_2-proj_{\vec{u}_1}\vec{v}_2$

Hence $\vec{u}_1.\vec{u}_2=0$

$\vec{u}_3=\vec{v}_3-proj_{\vec{u}_1}\vec{v}_3-proj_{\vec{u}_2}\vec{v}_3$

similarly $\vec{u}_3.\vec{u}_1=0$ *and* $\vec{u}_3.\vec{u}_2=0$

7. Ans. (A,B,C)

$$A=\begin{bmatrix}a & b & c\\ d & e & f\\ g & h & i\end{bmatrix}$$

$$|A-\lambda I|=0 \Rightarrow \begin{vmatrix}a-\lambda & b & c\\ d & e-\lambda & f\\ g & h & i-\lambda\end{vmatrix}=0$$

$$\Rightarrow \quad -\lambda^3+\lambda^2(a+e+i)$$

$$-\lambda(ae+ei+ia-gc-hf-bd)-|A|=0$$

Using Cayley Hamilton,

$$A^3-A^2\,tr(A)+\frac{(tr\,A)^2-trA^2}{2}A-|A|.I_3=0$$

$$p=tr(A),\ q=\frac{(tr\,A)^2-trA^2}{2}$$

$r = |A|$

8. Ans. (B,C,D)

Shortest distance

$$=\left|\frac{(-5\hat{i}+7\hat{j}-6\hat{k}).(\hat{i}-4\hat{j}+8\hat{k}|}{|\hat{i}-4\hat{j}+8\hat{k}|}\right|=9$$

9. Ans. (A,D)

$$\prod_{i=1}^{99}(x+i)=A_0+A_1x\ldots.+A_{99}x^{99}=\sum_{i=0}^{99}A_ix^i$$

Diff. w.r.t. 'x'

$$\prod_{i=1}^{99}(x+i).\sum_{i=1}^{99}\frac{1}{x+i}=\sum_{k=1}^{99}kA_r\,x^{k-1}$$

put x = 1

$$100!\sum_{i=1}^{99}\frac{1}{1+i}=\sum_{k=1}^{99}kA_r$$

so, $r_i=1+i$

$$S=\sum_{r=1}^{99}\frac{1}{\sqrt{r_k-1}}=\sum_{i=1}^{99}\frac{1}{\sqrt{i}}$$

Use $2\left(\sqrt{n+1}-\sqrt{n}\right)<\frac{1}{\sqrt{n}}<2\left(\sqrt{n}-\sqrt{n-1}\right)$

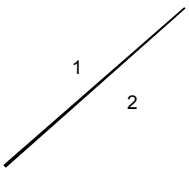
[S] = 18

10. Ans. (A,C)

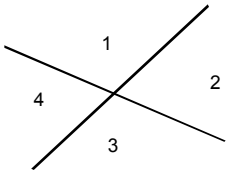
$$\text{Volume}=\left|\frac{(4-1)(6-3)(12+1)}{\begin{vmatrix}2 & 2 & 1\\ 2 & -1 & 1\\ 3 & 4 & 12\end{vmatrix}}\right|=\frac{13}{7}$$

11. Ans. (A,B,C,D)

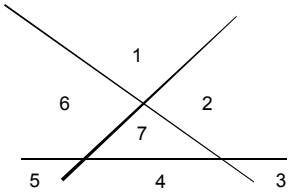
Clearly the number of regions is maximized when no two lines are parallel and no three and concurrent. Easy to see that $L_1 = 2$, $L_2 = 4$ and $L_3 = 7$. For the general case, we observe that when we add the nth line, this new line is cut by the previous $n-1$ lines in $n-1$ points and hence the new line is divided into n segments. Each of these segments divides a previous region into 2 parts and hence the number of new regions



$L_1 = 2$



$L_2 = 4$



$L_3 = 7$

Lines dividing a plane

added when we introduce the nth line is n. Thus we have the recurrence relation $L_n = L_{n-1} + n$. Hence $L_4 = 11, L_5 = 16, \ldots$.

$L_n - L_{n-1} = n$

$L_{n-1} - L_{n-2} = n-1$

$\ldots\ldots\ldots$

$L_3 - L_2 = 4$

$L_2 - L_1 = 2$

Adding, we get

$L_n - L_1 = n + (n-1) + (n-2) + \ldots + 2$

$= \frac{n(n+1)}{2} - 1$

Thus $L_n = \frac{n(n+1)}{2} - 1 + 2$

$L_{10} = 56$

$= \frac{n(n+1)}{2} + 1$

$L_{15} = 121$

$= \frac{1}{2}(n^2 + n + 2)$

$L_{20} = 211$

$L_{25} = 326$

12. Ans. (D)

$A(z_1)$ gives (1, 0); $B(z_2)$ gives $(2, \sqrt{3})$ & $C(z_3)$ is (5, 0)

$\therefore$ ar.(ΔABC) = $2\sqrt{3}$

& $AB = |z_1 - z_2| = \sqrt{1+3} = 2$

13. Ans. (A,C)

$19x^2 + 99s + 1 = 0$ roots are s and $\frac{1}{t}$

$\frac{s}{t} = \frac{1}{19}$

$s + \frac{1}{t} = -\frac{99}{19}$

$\frac{st+1}{t} = -\frac{99}{19}$

so, $\left|\frac{st+4s+1}{t}\right| = \left|\frac{-99}{19} + \frac{4}{19}\right| = 5$

SECTION-IV

1. Ans. 6

For consistency,

$$\begin{vmatrix} a & 1 & b \\ b & 1 & a \\ a & b & ab \end{vmatrix} = 0$$

$\Rightarrow$ $(a-b)(a-b^2) = 0$

$\therefore$ Possible ordered pairs are (0, 0), (1, 1), (2, 2), (3, 3), (4, 4), (4, 2)

2. Ans. 6

AB = BA

a = 2, b = 1

Now, $AB = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix} = 6I$

Now, $ABB^{-1} = 6B^{-1}$ $\Rightarrow$ $A = 6B^{-1}$

$\Rightarrow$ $A^{-1}AB = 6A^{-1}$ $\Rightarrow$ $B = 6A^{-1}$

$\therefore$ $(A + B) = 6(A^{-1} + B^{-1})$ $\Rightarrow$ $\lambda = 6$

3. Ans. 3

Vertex is (2, 2) and $e = \sqrt{1 + \frac{8}{16}} = \sqrt{\frac{3}{2}}$

so, focus are $(\pm\sqrt{6}, \pm\sqrt{6})$

4. Ans. 8

$z + \omega = i$ & $z^2 + \omega^2 = 1$

$\Rightarrow$ $z = -\frac{\sqrt{3}}{2} + \frac{i}{2}; \omega = \frac{\sqrt{3}}{2} + \frac{i}{2}$

$\therefore$ area of triangle $\Rightarrow$ $A = \frac{\sqrt{3}}{4}$

So, $2\sqrt{80A^2 + 1} = 2\sqrt{80\left(\frac{3}{16}\right) + 1} = 8$

5. Ans. 9

$f(x) = -\int_0^1 f^2(t)dt - 2x\int_0^1 f(t)dt$

$\therefore$ $f'(x) = -2\int_0^1 f(t)dt$ .....(i)

= constant

$\Rightarrow$ f must be linear.

So, let $f(x) = (ax + b)$

$\therefore$ $a = -2\int_0^1 (at + b)dt$

& $b = -\int_0^1 (at + b)^2 dt$

$\Rightarrow$ a = 3, b = –3

$\Rightarrow$ $f(x) = 3(x - 1)$

So, $f(4) = 3(4 - 1) = 9$

Paper Code : 1001CJA102117077

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2021 - 2022)

JEE (Main + Advanced) : ENTHUSIAST COURSE
SCORE-I

Test Type : FULL SYLLABUS Test Pattern : JEE-Advanced

TEST DATE : 16 - 06 - 2021

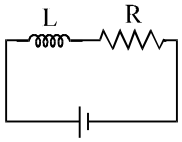
PAPER-2

PART-1 : PHYSICS SOLUTION

SECTION-I

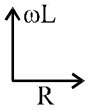
1. Ans. (B)

Sol. $Z_c = L/R$



$$\cos\phi = \frac{R}{\sqrt{R^2+\omega^2L^2}}$$

$$= \frac{1}{\sqrt{1+\omega^2\frac{L^2}{R^2}}} = \frac{1}{\sqrt{1+\omega^2\tau^2}}$$



2. Ans. (A)

Sol. $\Delta X = \frac{yd}{D}$ for central maxima $\Delta X = 0$, hence $y = 0$. (Does not depend upon λ)

$$B = \frac{\lambda D}{d}$$

$$\frac{n\lambda_1 D}{d} = \frac{m\lambda_2 D}{d}$$

$\Rightarrow$ $n \times 4500 = m \times 6000$

$3n = 4m.$

$\therefore$ $n = 4,\quad m = 3.$

3 Ans. (D)

Sol. (a) $\frac{1}{10} = \left(\frac{3}{2}-1\right)\left(\frac{1}{R}+\frac{1}{R}\right)$ $\therefore$ $R = 10$ cm

Now applying

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2-\mu_1}{R} \text{ twice}$$

We have, $\frac{3/2}{v_1} - \frac{1}{-20} = \frac{3/2-1}{10}$

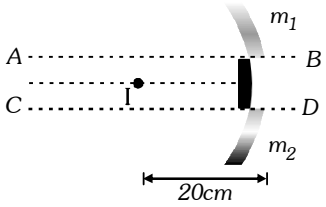
$$\frac{4/3}{v_2} - \frac{3/2}{v_1} = \frac{4/2-3/2}{-10}$$

Adding Eq. (i) and Eq. (ii), we hav $v_2 = 80$ cm

(b) The image formed by (lens + water) system will act as an object for the mirror. This is below the axis of m_1 and at the same distance as the centre is, therefore, its image will be formed vertically above at 1 mm from AB. Similarly m_2 will form an image of I_1 1mm below CD.

$I_1I_2 = 1 + 1 + 1 + 1 = 4$ mm

(c) $d = I_1I_2 = 4$ mm, $D = 80$ cm



$$\lambda = \frac{c}{\mu_w f} = \left(\frac{3.0\times10^8}{7.5\times10^{14}}\times\frac{3}{4}\right)$$

$$\omega = \frac{\lambda D}{d} = \frac{3}{4}\left(\frac{3.0\times10^8}{7.5\times10^{14}}\right)\left(\frac{80\times10^{-2}}{4\times10^{-3}}\right)m = 6\times10^{-5}m$$

$= 60\ \mu m$

4. Ans. (C)

Sol: $\phi = Li = BA$

$$i_{max} = \frac{B_{max}A}{L};$$

$$\frac{0.1x10^{-2}}{10x10^{-3}} = 0.1A = 100mA]$$

5. Ans. (C)

Sol. $T = 2\pi\left(\frac{a^3}{GM}\right)^{1/2}$ (Kepler's law)

and $a = \frac{r+R}{2}$

Hence, $T = \pi\sqrt{\frac{(r+R)^3}{2GM}}$

6. Ans. (D)

Sol. Plates are brought closer capacity will increase. As battery is removed charge remain constant. $U = \frac{1}{2}\frac{Q^2}{C}$

$\Rightarrow U \propto 1/C$. Hence stored energy will decrease.

7. Ans. (A, B)

8. Ans. (A,B,C)

9. Ans. (B,C)

Sol. $\frac{v_A}{v_B} = \sqrt{\frac{\gamma_A}{\gamma_B}\frac{M_B}{M_A}} = \sqrt{\frac{7\times3}{5\times5}\times\frac{4}{2}} = \sqrt{\frac{42}{25}}$

(M_A and M_B are molecular weight of A and B respectively)

$$\frac{3v_A}{2l_A} = \frac{3v_B}{4l_B} \Rightarrow \frac{l_A}{l_B} = \frac{2v_A}{v_B} = \sqrt{\frac{168}{25}}$$

10. Ans. (C,D)

Sol. Minimum distance from midpoint.

11. Ans. (A,C)

12. Ans. (A,B)

13. Ans. (B,C)

Sol. (A) $T = 2\pi\sqrt{\frac{\ell}{g_{eff}}}$

(B) fraction = $\frac{P_b}{P_L}$ same

(C) $T = 2\pi\sqrt{\frac{m}{k}}$

(D) $P = P_0 + hPg_{eff}$

14. Ans. (B,C)

Sol. (a) V leads I by $\pi/4$ inductive (R–L)

$R = \omega L \Rightarrow \omega = \frac{R}{L}$

(b) V lags I by $\pi/4$ capacitive (R–C)

$R = \frac{1}{\omega C} \Rightarrow \omega = \frac{1}{RC}$

(c) V & I are in same phase $X_L = X_C$

$\Rightarrow \omega = \frac{1}{\sqrt{LC}}$

(d) Not possible in R, L, C

15. Ans. (A)

16. Ans. (A)

Sol. $-q_0 + q_b - q_d + q_f = 0$ ....(1)

$V_{II} = 0$

$$\frac{Kq_0}{R_2} - \frac{Kq_0}{R_2} + \frac{Kq_b}{R_2} - \frac{Kq_b}{R_2} + \frac{Kq_d}{R_2} - \frac{Kq_d}{R_3} + \frac{Kq_f}{R_3} = 0$$

$$q_f = -q_d\left(\frac{R_3 - R_2}{R_2}\right) \quad ....(2)$$

$V_I = V_{III}$

$$\frac{Kq_0}{R} - \frac{Kq_0}{R} + \frac{Kq_b}{R} - \frac{Kq_0}{2R} + \frac{Kq_d}{2R} - \frac{Kq_d}{3R} + \frac{Kq_f}{3R}$$

$$= \frac{Kq_0}{3R} - \frac{Kq_0}{3R} + \frac{Kq_b}{3R} - \frac{Kq_0}{3R} - \frac{Kq_b}{3R} + \frac{Kq_d}{3R} - \frac{Kq_d}{3R} + \frac{Kq_f}{3r}$$

$q_d = -3q_b$ ....(3)

Solving eqn. $q_b = \left(\frac{2q}{11}\right)$

17. Ans. (C)

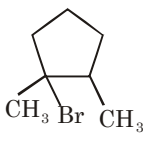
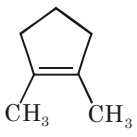
Sol. Mirror create the image at 0, –2 mm. Then TIR occur at the interface and the final image is created at 2, 0 mm.

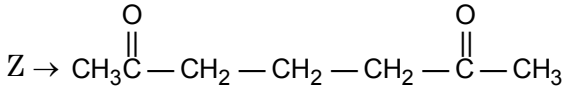
18. Ans. (B)

PART–2 : CHEMISTRY SOLUTION

1. Ans. (D)

2. Ans. (B)

Sol. X →  ; Y →  ;

Z → 

3. Ans. (C)

Sol. Disaccharides having hemiacetal group are reducing.

4. Ans. (C)

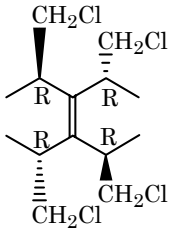
Sol. EA of H is -72 kJ/mol.

$\frac{1}{2}IP + \frac{1}{2}EA = 619.5$

$\frac{1}{2}IP - 36 = 619.5$

IP = 1311 kJ/mol

5. Ans. (C)

Sol.  No plane of symmetry.

Optically active but having C_2 axis of symmetry.

Presence of AoS doesn't matter for O. Activity.

6. Ans. (A)

Sol. nM = Mn

1 —

$1-\beta$ β/n

$$i = 0.9 = \frac{1-\beta+\beta/n}{1}$$

$$0.2 = \frac{\beta/n}{1-\beta+\beta/n} = \frac{\beta/n}{0.9} \Rightarrow \beta = 0.18n$$

$$0.9 = 1 - 0.18\,n + \frac{0.18n}{n}$$

$0.18n = 0.28$

$n = 28/18$

7. Ans. (A,C,D)

Sol.

$$\underset{\text{(excess)}}{6AgNO_3} + 3I_2 + 3H_2O \longrightarrow AgIO_3 + 5AgI + 6HNO_3$$

Hence choices (A), (C) and (D) are correct.

When excess of I_2 is added to silver nitrate solution then the reaction taking place can be represented as

$$5AgNO_3 + \underset{\text{(excess)}}{3I_2} + 3H_2O \longrightarrow HIO_3 + 5AgI + 5HNO_3$$

8. Ans. (A,B)

Sol. Choices (A) and (B) are correct because in extraction of copper SiO_2 is used as impurity.

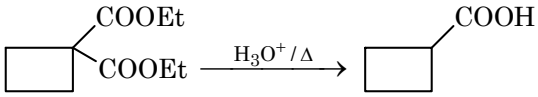
$$\underset{\text{Ferrous oxide}}{FeO} + \underset{\text{Silica}}{SiO_2} \longrightarrow \underset{\text{Ferrous silicate}}{FeSiO_3}$$

(B) is correct because limestone is used to remove impurities in extraction of iron.

$$\underset{\substack{\text{Calcium}\\\text{carbonate}\\\text{(flux)}}}{CaCO_3} + \underset{\substack{\text{Silica}\\\text{(impurity)}}}{SiO_2} \longrightarrow \underset{\substack{\text{Calcium Silicate}\\\text{(slag)}}}{CaSiO_3} + CO_2$$

9. Ans. (A,B)

Sol. $CH_2(COOEt)_2 + (CH_2)_3Br_2 \xrightarrow[OEt]{S}$



10. Ans. (B,C,D)

Sol. $Zn + Cu^{+2} \longrightarrow Zn^{+2} + Cu$

– 0.9 M 0.1 M –

– 0.5 M 0.5 M –

0.8 M 0.2 M

$$E = 1.1 - \frac{0.06}{2}\log\frac{1}{4}$$

$$= 1.1 + \frac{0.06}{2} \times 2 \times 0.30 = 1.118\ V$$

11. Ans. (A,B,C)

Sol. (A) Rate = K [Conc.]^0

$$\text{Rate} = Ae^{-\frac{Ea}{RT}}$$

Unit of A will be same as of rate of reaction

(B) $K_1 = \frac{1}{30}\ell n\left[\frac{100}{75}\right]$ $K_2 = \frac{1}{60}\ell n\left[\frac{100}{50}\right]$

$K_1 \neq K_2$

(C) $K_1 = \frac{1}{30}\left[\frac{1}{2a/3} - \frac{1}{a}\right]$ $K_2 = \frac{1}{90}\left[\frac{1}{a/3} - \frac{1}{a}\right]$

$K_1 = \frac{1}{30} \times \frac{1}{a}[.5]$ $K_2 = \frac{2}{90 \times a}$

$K_1 \neq K_2$

(D) Zero order

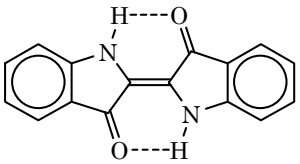
$K_1 = \frac{a - 4a/5}{30}$ $K_2 = \frac{a - 2/5a}{90}$

$K_1 = \frac{a}{150}$ $K_2 = 3/5\frac{a}{90}$

$K_2 = \frac{a}{150}$

$K_1 = K_2$

12. Ans. (A,B,D)

Sol. 

$\Rightarrow$ Can show GI

$\Rightarrow$ Trans is more stable than cis due to H–bond.

$\Rightarrow$ It is planer.

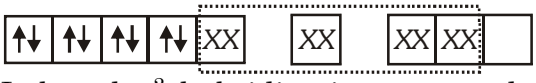
13. Ans. (A,B,C)

Sol. Electron withdrawing group in phenol decreases rate of reaction.

14. Ans. (B,C)

Sol. (B) is correct Pt(78) $[Xe]_{54}\ 4f^{14}\ 6s^1\ 5d^9$

$Pt^{2+}\ 6s^0\ 5d^8$

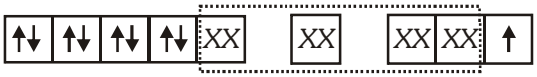


It has dsp^2 hybridisatio, square planar shape.

(C) is correct Cu(29) $[Ar]\ 4s^1\ 3d^{10}$

$Cu^{2+}\ 4s^0\ 3d^9$

$[Cu(NH_3)_4]^{2+}$



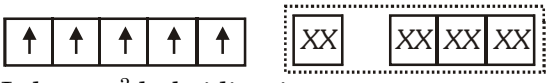
It has dsp^2 hybridisation, square planar shape.

(A) is not correct, has 'Ni' in sp^3 hybridised and tetrahedral in shape.

(D) is not correct, $[FeCl_4]^-$ has sp^3 hybridisation and tetrahedral shape because Cl^- is weak field ligand, cannot cause pairing of electrons.

Fe $[Ar]\ 4s^2 3d^6$

$Fe^{3+}\ 4s^0 3d^5$

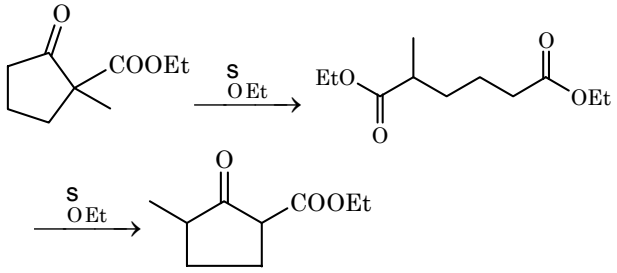


It has sp^3 hybridization.

15. Ans. (C)

16. Ans. (C)

Sol. (15 to 16)



17. Ans. (B)

18 Ans. (D)

Sol. (17 to 18)

$$\underset{(P)}{CuSO_4} \xrightarrow{BaCl_2} BaSO_4 \downarrow (\text{white ppt})$$

$$CuSO_4 + \underset{\text{excess}}{NH_3} \longrightarrow \underset{(Q)}{[Cu(NH_3)_4]SO_4}[\text{deep blue}]$$

$$CuSO_4 + KCN \longrightarrow Cu(CN)_2 \downarrow \text{yellow ppt}$$

$$2Cu(CN)_2 \longrightarrow \underset{(R)}{2CuCN} + (CN)_2[\text{Cynogen highly poisonous}]$$

$$(CN)_2 \xrightarrow{Pt/H_2} NH_2\text{—}CH_2\text{—}CH_2\text{—}NH_2\ (S)$$

$$CoCl_3 + 3en \longrightarrow \underset{(T)}{[Co(en)_3]Cl_3}$$

S is bidentate ligand & R is cynogen.

$\therefore$ Option (B) is correct.

Q is $[Cu(NH_3)_4]SO_4$ & T is $[Co(en)_3]Cl_3$

Hence A, B and C all the options are correct.

$\therefore$ (D) is the correct options.

PART-3 : MATHEMATICS — SOLUTION

SECTION-I

1. Ans. (D)

$$\ln(x-a)+\ln(x-b)+\ln(x-c)+\ln(x-d)=\ln(x^4+5x^3-4x^2+6)$$

$$\frac{1}{x-a}+\frac{1}{x-b}+\frac{1}{x-c}+\frac{1}{x-d}=\frac{4x^3+15x^2-8x}{x^4+5x^3-4x^2+6}$$

Let $\sum_{cyc}\frac{1}{x-a}=g(x)$

$$\sum_{cyc}\frac{(3-a)-(2-a)}{(2-a)(3-a)}=g(2)-g(3)$$

$$\sum_{cyc}\frac{1}{a^2-5a+6}=\frac{38}{23}-\frac{219}{186}=\frac{677}{1426}$$

2. Ans. (C)

Let a_n denote the number of ways in which a 2 × n rectangle be filled with tiles of size 1 × 2. Clearly, $a_1 = 1$ and $a_2 = 2$. For $n \geq 3$, we will build the titles one tile at a time. If the first tile is vertical, then we have $2\times(n-1)$ rectangle that can be tiles in a_{n-1} ways. If the first tile is horizontal, then the next tile is also horizontal and we are left with $2\times(n-2)$ rectangle and this can be tiled in a_{n-2} ways. With the initial

conditions $a_1 = 1$ and $a_2 = 2$, this yields the sequence 1, 2, 3, 5, 8, 13, 21, 34, 55, 89

$\therefore \quad a_{10} = 89$

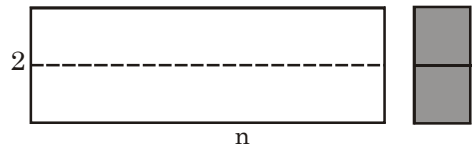


Figure-1 (Tilling a $2 \times n$ rectangle)

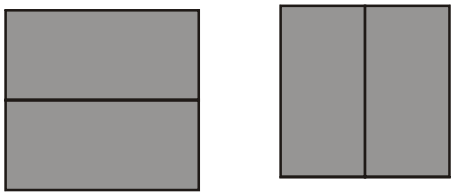


Figure-2 (Tilling a 2×2 rectangle)

3. Ans. (D)

Let $a = \sin^2 x$ and $b = \cos^2 x$. Since $\frac{1}{2}\left(\frac{1}{a}+\frac{1}{b}\right) \geq \left(\frac{2}{a+b}\right)$ and $a+b=1$ we have

$$\left(\frac{1}{a}+\frac{1}{b}\right) \geq 4$$

where equality holds only when $a = b = 1/2$. Therefore

$$12 + \frac{1}{2}\sin y = \left(a+\frac{1}{a}\right)^2 + \left(b+\frac{1}{b}\right)^2$$

$$\geq \frac{1}{2}\left\{\left(a+\frac{1}{a}\right)+\left(b+\frac{1}{b}\right)\right\}^2$$

$$= \frac{1}{2}\left(a+b+\frac{1}{a}+\frac{1}{b}\right)^2 \geq \frac{1}{2}(1+4)^2 = \frac{25}{2}$$

with equality only when $a = b = 1\frac{1}{2}$. So we have $\sin y \geq 1$; hence sin y = 1; this also implies $a = b = \frac{1}{2}$;

therefore $\sin^2 x = \frac{1}{2}$; i.e., $\sin x = \pm\frac{1}{\sqrt{2}}$ Thus the solutions are

$$x = \frac{\pi}{4} + \frac{k\pi}{2}, \quad y = \frac{\pi}{2} + 2l\pi, \quad k,l \in \mathbb{Z}.$$

4. Ans. (B)

$z^7 - 7 = (z-z_0)(z-z_1)....(z-z_6)$

Putting (1) $z = -z_0$

$$-14 = -z_0^7 - 7 = -2z_0(z_0+z_1)(z_0+z_2).....(z_0+z_6)$$

(2) $z = -z_1$

$-14 =$

$$-z_1^7 - 7 = -2z_1(z_1+z_0)(z_1+z_2).....(z_1+z_6)$$

and similarly $z = -z_2, -z_3, ..., -z_6$

Multiplying all such equations

$$-14^7 = -2^7(z_0 z_1 ....z_6)\left(\prod_{0\leq i\ \leq j\leq 6}(z_i+z_j)\right)^2$$

$$7^6 = \prod_{0\leq i\leq j\leq 6}(z_i+z_j)^2 = p^2$$

5. Ans. (C)

Let $S = \sum_{n=1}^{99}\sqrt{10+\sqrt{n}}$, $T = \sum_{n=1}^{99}\sqrt{10-\sqrt{n}}$. Note that $\sqrt{a+b+2\sqrt{ab}} = \sqrt{a}+\sqrt{b}$ for positive real numbers a and b. In particular, for each fixed positive integer n with $1 \leq n \leq 99$, choose $a > b > 0$ satisfying $a+b=20$, $ab=n$, then a, b are roots of the equations $x^2 - 20 + n = 0$ i.e.

$a = 10+\sqrt{100-n}$, $\quad b = 10-\sqrt{100-n}$

$$\text{so } \sqrt{2}S = \sum_{n=1}^{99}\sqrt{20+2\sqrt{n}} = \sum_{n=1}^{99}\left(\sqrt{10+\sqrt{100-n}}+\sqrt{10-\sqrt{100-n}}\right)$$

$$= \sum_{n=1}^{99}\left(\sqrt{10+\sqrt{n}}+\sqrt{10-\sqrt{n}}\right) = S+T$$

Thus, $\left(\sqrt{2}-1\right)S = T \Rightarrow \frac{S}{T} = \frac{1}{\sqrt{2}-1} = 1+\sqrt{2}$.

6. Ans. (B)

X = {1 2, 3, ... n}

Sample space

$A \cap C = \phi$

$B \cap D = \phi$

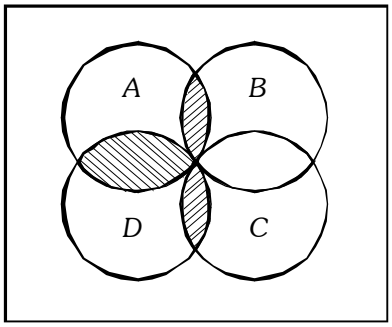
Events

$E_1 : A \cap B = \phi$; $E_2 : B \cap C = \phi$;

$E_3 : C \cap D = \phi$; $E_4 : D \cap A = \phi$

$\bar{E}_1 : A \cap B \neq \phi$; $\bar{E}_2 : B \cap C \neq \phi$;

$\bar{E}_3 : C \cap D \neq \phi$; $\bar{E}_4 : D \cap A \neq \phi$



$n(S) = 9^n$

$n(E) = n(\bar{E}_1 \cap \bar{E}_2 \cap \bar{E}_3 \cap \bar{E}_4)$

$= n\left(\overline{E_1 \cup E_2 \cup E_3 \cup E_4}\right)$

$= 9^n - \left(E_1 \cup E_2 \cup E_3 \cup E_4\right)$

$[(\Sigma n(E_1) - \Sigma n(E_1 \cap E_2) + \Sigma n(E_1 \cap E_2 \cap E_3)]$

$= 9^n - [4.8^n - 6.7^n + 4.6^n - 5^n]$

7. Ans. (A,B,C)

$|\text{adj}(\text{adj}(\text{adj}(\text{adjA})))| = |A|^{16} = 4^8 . 5^{16}$

$\Rightarrow |A| = 10$

$\Rightarrow x + y + z = 10$

So p = 36

As we know that

$$\int_a^b f(x)dx + \int_c^b f^{-1}(x)dx = bd - ac,$$

where $f(x)$ is increasing and invertible function.

Also $f(b) = d, f(a) = c$.

Now,

$g(x) = 2 + 2^{\sin x}$, then $g^{-1}(x) = \sin^{-1}(\log_2(x-2))$

Hence, $\int_{-\pi/2}^{\pi/2}(2+2^{\sin x})dx + \int_{5/2}^{4}\sin^{-1}(\log_2(x-2))dx = 2\pi + \frac{5\pi}{4}$

$\Rightarrow 2\pi + \int_{-\pi/2}^{\pi/2}(2^{\sin x})dx + \int_{5/2}^{4}\sin^{-1}(\log_2(x-2))dx = 2\pi + \frac{5\pi}{4}$

$\Rightarrow \int_{-\pi/2}^{\pi/2}(2^{\sin x})dx + \int_{5/2}^{4}\sin^{-1}(\log_2(x-2))dx = \frac{5\pi}{4}$

Hence q = 5

Now,

$[\vec{a}\times\vec{b}\ \ \vec{b}\times\vec{c}\ \ \vec{c}\times\vec{a}] = [\vec{a}\vec{b}\vec{c}]^2 = \left|\vec{c}\cdot(\vec{a}\times\vec{b})\right|^2 = |\vec{c}|^2|\vec{a}\times\vec{b}|^2\cos^2\theta$

The maximum value of $[\vec{a}\vec{b}\vec{c}]^2$ is possible only when $\vec{c}$ is parallel to $\vec{a}\times\vec{b}$

$\Rightarrow \theta = 0$ or π

Hence, $\left|(\vec{a}\times\vec{b})\times\vec{c}\right|^2 = |\vec{c}|^2|\vec{a}\times\vec{b}|^2\sin^2\theta = 0$

8. Ans. (A,C)

Effective parameters = 1 = order

$$\frac{dy}{dx} = k\frac{(e^{\sqrt{x}} - e^{-\sqrt{x}})}{2\sqrt{x}}$$

$$\frac{dy}{dx} = \frac{(e^{\sqrt{x}} - e^{-\sqrt{x}})}{2\sqrt{x}} \cdot \frac{y}{(e^{\sqrt{x}} + e^{-\sqrt{x}})}$$

degree = 1

9. Ans. (A,D)

$$\lim_{x\to 0}\frac{x^{6000} - (\sin x)^{6000}}{x^2(\sin x)^{6000}} = \lim_{x\to 0}\frac{\left(\frac{x}{\sin x}\right)^{6000} - 1}{x^2}$$

$$= \lim_{x\to 0} 6000 \times \frac{x - \sin x}{x^2 \sin x} = 6000 \times \frac{1}{6} = 1000$$

for $= \lim_{x\to 0}\frac{1 - \cos^5 x \cos^3 2x \cos^3 3x}{x^2}$ (use L.H. Rule)

10. Ans. (B,C)

Since x = –1 and x = 0 are the two critical values of the absolute functions, one can suppose that $F(x) = a|x+1| + b|x| + cx + d$

$$= \begin{cases} (c-a-b)x + d - a & \text{if } x < -1 \\ (a+c-b)x + a + d & \text{if } x - 1 \le x \le 0 \\ (a+b+c)x + a + d & \text{if } x > 0 \end{cases}$$

which implies that a = 3/2, b = –5/2, c = –1 and d = 1/2.

Hence $f(x) = (3x+3)/2, g(x) = 5x/2$, and $h(x) = -x + \frac{1}{2}$

11. Ans. (A,B,C)

Putting $x = \frac{1}{2}$ will give m = 170.

Hence, four distinct roots $\frac{1}{8}, \frac{1}{2}, 2, 8$ which form a geometric progression with the ratio r = 4

12. Ans. (A,C)

Let n be the number of the pages of the book. Suppose the leaf torn bears the pge numbers k and k + 1. Then

$$\frac{n(n+1)}{2} = 15000 + k + (k+1)$$

Therefore

$n(n+1) = 30000 + 4k + 2$ (...1)

Since $k \ge 1$, we have

$n(n+1) \ge 30006 > 172 \cdot 173 \Rightarrow n \ge 173$

Since $k \le n$,

$n(n+1) \le 30000 + 4n + 2$;

i.e., $n^2 - 3n + 2 \le 30004$

$\Rightarrow \quad (n-2)(n-1) < 173 \cdot 174$

$\Rightarrow \quad n - 2 < 173$

$\Rightarrow \quad n \le 174$

Note that $n \ne 174$, for otherwise substituting $n = 174$ in (1), we would get $k = 112$, a contradiction because k being a right-page number, should be odd. So taking $n = 173$, we get from (1), that $k = 25$; hence the page numbers of the torn leaf are 25 and 26.

13. Ans. (B,D)

Pankaj has the winning strategy. A set of three distinct rational nonzero numbers a, b and c, such that $a + b + c = 0$, will do the trick. Let A, B and C be any arrangement of a, b and c, and let $f(x) = Ax^2 + Bx + C$.

Then $f(x) = Ax^2 + Bx + C$.

which implies that 1 is a solution.
Since the product of the two solutions is $\frac{C}{A}$, the other solution is $\frac{C}{A}$, and it is different from 1.

14. Ans. (B,C)

Rearranging terms by power of 2 yields

$$2^{x^2}x + 2^{x+1}(x^2-1) - 2(x^2+x-1) = 0 \quad ...(1)$$

Setting $y = x^2 - 1$ and dividing by 2 on the both sides, (1) becomes

$$2^y x + 2^x y - (x+y) = 0$$

or $\quad x(2^y - 1) + y(2^x - 1) = 0 \qquad ...(2)$

Since $f(x) = 2^x - 1$ and x always have the same sign,

$$x(2^y - 1) \cdot y(2^x - 1) \ge 0$$

Hence if the terms on the left-hand side of (2) are nonzero, they must have the same sign, which in term implies that their sum is not equal to 0.
Therefore (2) is true if and only if
$x = 0$ or $y = 0$, which leads to solutions
$x = -1,\ 0$ and 1

15. Ans. (D)

$$\frac{\Gamma\left(\frac{10+1}{2}\right)\Gamma\left(\frac{12+1}{2}\right)}{2.\Gamma\frac{10+12+2}{2}} = \frac{\left(\frac{9}{2}\cdot\frac{7}{2}\cdot\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\sqrt{\pi}\right)\cdot\left(\frac{11}{2}\cdot\frac{9}{2}\cdot\frac{7}{2}\cdot\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\sqrt{\pi}\right)}{12!}$$

16. Ans. (A)

$$I_b = \frac{b!}{a^{b+1}} = \frac{\Gamma(b+1)}{a^b + 1}$$

$$\int_0^\infty t^b e^{-at}\,dt = \left(t^b \frac{e^{-at}}{a}\right)_0^\infty + \frac{b}{a}\int_0^\infty t^{b-1}\, e^{-at}\, dt$$

$$I_b = \frac{b}{a}I_{b-1} = \frac{b(b-1)}{a^2}\, I_b - 2 \,.....= \frac{b!}{a^b}\, I_0$$

$$I_b = \int_0^\infty e^{-at}\,dt = \frac{1}{a}$$

$$I_b = \frac{b!}{a^{b+1}} = \frac{\Gamma(b+1)}{a^b + 1}$$

17. Ans. (D)

For a fixed positive integer n in range $\left[n, n + \frac{1}{n}\right)$, $f(x) = n\left|x - n - \frac{1}{2n}\right| = \left|nx - n^2 - \frac{1}{2}\right|$

for example when $n = 4, f(x) = \left|4x - 16 - \frac{1}{2}\right|$

for

$$4 \le x < 4\frac{1}{4}, f(4) = f\left(4\frac{1}{4}\right) = \frac{1}{2} \text{ and } f\left(4\frac{1}{8}\right) = 0$$

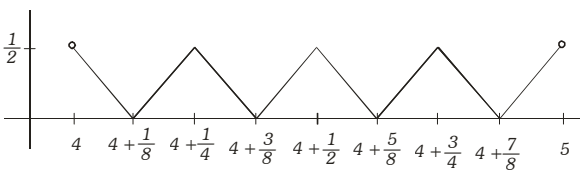
In general

$$f(n) = f\left(n + \frac{1}{n}\right) = \frac{1}{2} \text{ and } f\left(n + \frac{1}{2n}\right) = 0$$

Thus graph of f on $\left[n, n + \frac{1}{n}\right)$ is like V shape

18. Ans. (B)

For $x \in (4,5)$ graph is



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(Academic Session : 2021 - 2022)

JEE (Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)

ANSWER KEY : PAPER-1 **TEST DATE : 16-06-2021**

Test Type : FULL SYLLABUS Test Pattern : JEE-Advanced

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	D	C	B	B,C,D	B,D	A,C	A,C	B,C
	Q.	11	12	13							
	A.	A,B	A,C,D	A,B,C,D							
SECTION-IV	Q.	1	2	3	4	5					
	A.	6	6	3	5	5					

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	C	D	D	B	A,B,C	A,B,C	C,D	B,C	A,C
	Q.	11	12	13							
	A.	A,B,D	A,B,C,D	A,C,D							
SECTION-IV	Q.	1	2	3	4	5					
	A.	3	4	8	6	7					

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	A	B	C	D	A,C,D	A,B,C	B,C,D	A,D	A,C
	Q.	11	12	13							
	A.	A,B,C,D	D	A,C							
SECTION-IV	Q.	1	2	3	4	5					
	A.	6	6	3	8	9					

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	Q.	11	12	13							
	A.	A,B	A,C,D	A,B,C,D							
SECTION-IV	Q.	1	2	3	4	5					
	A.	6	6	3	5	5					

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	A.	D	C	D	D	B	A,B,C	A,B,C	C,D	B,C	A,C
	Q.	11	12	13							
	A.	A,B,D	A,B,C,D	A,C,D							
SECTION-IV	Q.	1	2	3	4	5					
	A.	3	4	8	6	7					

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	A	B	C	D	A,C,D	A,B,C	B,C,D	A,D	A,C
	Q.	11	12	13							
	A.	A,B,C,D	D	A,C							
SECTION-IV	Q.	1	2	3	4	5					
	A.	6	6	3	8	9					

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Test Type : FULL SYLLABUS Test Pattern : JEE-Advanced

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	D	C	C	D	A,B	A,B,C	B,C	C,D
	Q.	11	12	13	14	15	16	17	18		
	A.	A,C	A,B	B,C	B,C	A	A	C	B		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	C	C	C	A	A,C,D	A,B	A,B	B,C,D
	Q.	11	12	13	14	15	16	17	18		
	A.	A,B,C	A,B,D	A,B,C	B,C	C	C	B	D		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	C	D	B	C	B	A,B,C	A,C	A,D	B,C
	Q.	11	12	13	14	15	16	17	18		
	A.	A,B,C	A,C	B,D	B,C	D	A	D	B		

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PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	D	C	C	D	A,B	A,B,C	B,C	C,D
	Q.	11	12	13	14	15	16	17	18		
	A.	A,C	A,B	B,C	B,C	A	A	C	B		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	C	C	C	A	A,C,D	A,B	A,B	B,C,D
	Q.	11	12	13	14	15	16	17	18		
	A.	A,B,C	A,B,D	A,B,C	B,C	C	C	B	D		

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SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	C	D	B	C	B	A,B,C	A,C	A,D	B,C
	Q.	11	12	13	14	15	16	17	18		
	A.	A,B,C	A,C	B,D	B,C	D	A	D	B		